A computer-assisted lower bound for the no-three-in-line problem

A 481-phase construction attaining $1.501n$

10 November 2026

Version 0.2: the 481-phase construction

Abstract

We give a computer-assisted proof that, for every sufficiently large integer n, the $n \times n$ integer grid contains a set of at least $\lceil 1.501n \rceil$ points with no three collinear. The construction uses a finite continuous-time process whose birth rates compensate for geometric deletion and prescribed halvings. A saved-credit martingale identifies the expected output with deterministic input mass minus exposure lost at early retirement. Common-stopping mixed moments and protected-line concentration bound the five causes of this loss.

The finite input has 481 phases on a 256×256 spatial partition, four parity types, and exact mass 1.501114506104296875. Its capacity and root-deadline risk bounds are checked using integer and rational upper bounds over entire spatial cells and angular intervals. The accompanying package contains the input, checker source and full fresh replay records. Fixed moment orders and a total loss budget of 10^{-5} suffice for the final comparison. No divisibility restriction on n is used. The argument is asymptotic and does not provide a practical numerical threshold for n.

Reading and attribution. No knowledge of Lean is needed for the mathematical text. The probabilistic arguments are proved in ordinary mathematical language; the executable checks concern the finite geometric input. AI systems contributed substantially to the research, code, proofs, checking and exposition. No named author is supplied. This manuscript has not received independent human peer review, and the separate partial Lean development is not a formal proof of its final theorem. Precise disclosure and execution scope appear in Section 11.

Contents

1 The target and the structure of the argument **3**

1.1 Finite input, fixed before the grid size 3

1.2 The geometric input conditions 3

1.3 Why saved credit is the useful quantity 4

1.4 A fixed budget suffices for the numerical target 5

2 Uniform lattice geometry **5**

2.1 Parity, chords and a uniform boundary estimate 6

2.2 A finite sieve for primitive directions 7

2.3 The common-deadline pair sum 8

2.4 Explicit constants for the 481-phase input 9

3 The stopped compensated process 10
3.1 Line variables, rates and flows 10
3.2 Safety boundaries and the observation convention 11

4 Saved credit and the exact output identity 12

5 Mixed moments without independence 13

6 Removing the global safety stops 15
6.1 Pair-correction and virtual-deletion guards 16
6.2 The protected law of a designated line 17
6.3 Explicit clock constants for the verified input 19

7 Ordinary-line risk under the point law 21
7.1 Removing ordinary-line credit at additive cost 21
7.2 Long lines: replace the leaf sum, retain the birth time 22
7.3 Short stars and their exceptional configurations 23
7.4 Hazard concentration and the weight cap 24

8 Ordinary-line tags and the final loss estimate 24

9 The finite input and its geometric checker 26
9.1 A schedule defined by reverse deletion ranks 26
9.2 Exact maxima on a closed root cell 27
9.3 A rigorous angular-bin bound by vertical dilation 28
9.4 Complete Farey partitions and rational integral bounds 29
9.5 Four direction families, integer ranges and acceptance 30
9.6 Capacity: axes, finite directions and the infinite tail 30

10 The 481-phase application and a fixed error budget 32
10.1 One concrete finite input 32
10.2 Three global causes with fixed orders 33
10.3 The weight cap with a fixed moment order 33
10.4 Ordinary-line tags with a fixed moment order 34
10.5 The final rational comparison 34

11 Verification scope and provenance 35
11.1 Finite computation and mathematical proof 35
11.2 Companion files 35
11.3 AI use and licensing 36

1 The target and the structure of the argument

Write $V_n = \{0, \dots, n-1\}^2$. A set $S \subseteq V_n$ has the *no-three-in-line property* if every three distinct points of S are noncollinear. Equivalently,

$$(q_x - p_x)(r_y - p_y) - (q_y - p_y)(r_x - p_x) \neq 0 \qquad (p, q, r \in S \text{ distinct}).$$

Every such set has at most two points on each horizontal row, hence $|S| \leq 2n$. The numerical conclusion proved below is

$$\exists n_0 \; \forall n \geq n_0 \; \exists S \subseteq V_n : \quad \mathrm{NoThree}(S), \qquad 1501n \leq 1000|S|. \tag{1}$$

The positive improvement over $3n/2$ is the fixed number $n/1000$. The quantifier is over every sufficiently large integer n, with no divisibility restriction.

1.1 Finite input, fixed before the grid size

Choose a finite rational time partition $0 = t_0 < t_1 < \cdots < t_J = T$ and a fixed finite rational rectangular partition of the unit square. A common refinement may be taken to be a $G \times G$ grid for a fixed integer G. Let $\mathcal{B}$ be the union of its horizontal and vertical grid lines. There are four parity types $r \in (\mathbb{Z}/2\mathbb{Z})^2$. For each type choose a deterministic function

$$f_{r,t} : [0,1]^2 \longrightarrow [0,1]$$

that is constant on every open spatial cell and every time interval. Set it to zero on $\mathcal{B}$, outside the square, and after time T. Define

$$m_t(p) = f_{\mathrm{typ}(p),t}(p/n), \qquad F_{r,t}(x) = \int_0^t f_{r,u}(x)\,\mathrm{d}u, \qquad A = \frac{1}{4}\sum_r \int_{[0,1]^2} F_{r,T}(x)\,\mathrm{d}x. \tag{2}$$

Here $\mathrm{typ}(p) = p \bmod 2$. The factor $1/4$ belongs to the spatial mass, not to the birth rate of an individual point. The last-use time $\tau_r(x)$ is the right endpoint of the last interval on which that type and cell is used; it is zero for an unused cell. For a grid point write $\tau_p = \tau_{\mathrm{typ}(p)}(p/n)$.

The concrete construction has $G = 256$ and 481 phases. An integer field and a rank assigned to each positive type-cell specify its one exposure phase. Ranks are used in decreasing order, so the cumulative field at the end of the phase of rank j is supported on ranks at least j. Section 9 gives the exact masks, durations and data format. Removing $\mathcal{B}$ is part of the definition of the discrete process, even though it has zero planar area.

1.2 The geometric input conditions

A primitive integer direction v is identified with $-v$. Put $h(v) = \max(|v_1|, |v_2|)$ and $\bar{v} = v \bmod 2 \neq 0$. For a field F define the major-coordinate chord integral

$$C_v F(x) = \int_{\mathbb{R}} F\big(x + sv/h(v)\big)\,\mathrm{d}s.$$

This is not an arc-length integral. Let Γ be a fixed finite family of designated directions containing both coordinate directions, of cardinality g. The numerical example uses all directions of height at most three, so $g = 16$. Define the line capacity

$$L_{v,r}(F; x) = \frac{C_v(F_r + F_{r+\bar{v}})(x)}{2h(v)}. \tag{3}$$

Let the four direction families be $(1,u),(1,-u),(u,1),(-u,1)$, $0 \le u \le 1$, and let $c_{F,j}(x,u)$ denote the corresponding chord. Set

$$g_F(x) = \frac{1}{4}\sum_{j=1}^{4}\int_0^1 c_{F,j}(x,u)^2\,\mathrm{d}u, \qquad \mathcal{R}_r(F;x) = \frac{D_*}{6}\sum_{s\neq r} g_{F_r+F_s}(x). \tag{4}$$

Fix a finite set P of pairwise coprime odd integers greater than one and require $D_* \ge D_P = \frac{3}{4}\prod_{q\in P}(1-q^{-2})$. This is part of the admissibility hypothesis below. Section 2.2 proves the resulting finite-sieve bound; the concrete input uses $D_* = 76/125$ and 175 odd prime factors.

The finite schedule is *admissible* for fixed parameters if there are positive constants κ_H, κ_L and a number $0 \le r_* < d$ such that

$$L_{v,r}(F_T;x) \le K - \kappa_H \qquad (v \in \Gamma), \tag{5}$$
$$L_{v,r}(F_T;x) \le \lambda - \kappa_L \qquad (v \notin \Gamma), \tag{6}$$
$$\mathcal{R}_r(F_{\tau_r(x)};x) \le r_* \qquad (\tau_r(x) > 0), \tag{7}$$

uniformly over the relevant roots, types and lines. In the last formula *every one of the four fields is evaluated at the same root's deadline.* The constants satisfy

$$\begin{gathered} 0 < K < 2, \quad 0 < \rho < 1, \quad 0 \le \lambda < 1-\rho, \\ 0 < d < \gamma < 1, \quad 0 < \xi < d, \quad \zeta > 0, \quad 1 - d - 2\xi - \zeta > 0. \end{gathered} \tag{8}$$

All data and all displayed constants are fixed before n tends to infinity. Constants depending on a moment order may depend on that order, but the order is also fixed before taking the limit.

Theorem 1.1 (Analytic reduction)**.** *For an admissible finite schedule, the stopped compensated process defined in Section 3 produces a random set $S_n \subseteq V_n$ with the no-three-in-line property and*

$$\mathbb{E}|S_n| = (A - o(1))n.$$

Consequently, for every fixed $c < A$, a set of size at least cn exists for every sufficiently large n.

The detailed argument for this theorem occupies Sections 2–8. The concrete schedule and the exact finite verification of (5)–(7) are given in Sections 9 and 10. Corollary 10.1 combines them with the analytic reduction to obtain (1). The finite computation verifies the geometric input; the probabilistic lemmas are proved in the text.

1.3 Why saved credit is the useful quantity

Each active point has a weight w_p, initially one. It is selected at rate $m_t(p)w_p/n$. Weights grow to compensate for the chance of selection, deletion by a newly completed pair, or a prescribed halving. At an early safety stop, the positive weight is saved rather than discarded. The resulting process Z_p is a bounded martingale with $\mathbb{E}Z_p(t) = 1$.

The change of probability measure

$$\mathrm{d}\mathbb{P}^{(p)} = Z_p(T)\,\mathrm{d}\mathbb{P}$$

measures the amount of credit lost by retiring p, rather than just the raw probability that p retires. Let θ_p be its first positive saving time and E_p the event that this happens strictly before τ_p for a safety reason. Put

$$A_n = \frac{1}{n^2}\sum_{p\in V_n} F_{\mathrm{typ}(p),T}(p/n).$$

The exact identity proved in Section 4 is

$$A_n - \frac{\mathbb{E}|S_n|}{n} = \frac{1}{n^2}\sum_p \mathbb{E}^{(p)}\left[\mathbf{1}_{E_p}\int_{\theta_p}^{T} m_t(p)\,\mathrm{d}t\right]. \tag{9}$$

Thus a bound $\sup_p \mathbb{P}^{(p)}(E_p) \le \varepsilon$ implies $\mathbb{E}|S_n|/n \ge (1-\varepsilon)A_n$. There are five causes to control: weight cap, ordinary-line tag, pair-correction floor, virtual-deletion bound, and designated-line clock. The last three are global safety stops. The paper proves their estimates on the already stopped process; it does not assume that those stops are rare while deriving the estimates that make them rare.

1.4 A fixed budget suffices for the numerical target

The exact mass and target are

$$A = \frac{3842853135627}{2560000000000}, \qquad c = \frac{1501}{1000}, \qquad \Delta = A - c = \frac{293135627}{2560000000000}. \tag{10}$$

The grid comparison gives $A_n \ge A(1-512/n)^2$. For sufficiently large n, the five stopping-cause estimates with fixed parameters give $\sup_p \mathbb{P}^{(p)}(E_p) < 10^{-5}$. Taking also $n \ge 2^{24}$,

$$(1-10^{-5})A(1-2^{-15})^2 - c = \frac{2165079963272809251797}{27487790694400000000000000000} > 0.$$

This strict rational inequality is enough for (1). There is no need to prove an effective, practically small threshold n_0, or to compute an unbounded sequence of ever tighter error budgets. The asymptotic argument is a way to obtain one fixed budget. The condition $n \ge 2^{24}$ bounds only the displayed mass error; the probability estimates impose additional asymptotic thresholds.

2 Uniform lattice geometry

In this section a line means its intersection with V_n, and only lines containing at least two grid points are counted. Write L_ℓ for its cardinality and L_{pq} for the length of the line through distinct p, q. Constants in this section are uniform in the root's position, including roots approaching a cell boundary as n changes.

Lemma 2.1 (Direction and line counts)**.** *There are at most $4h$ primitive unoriented directions of height h. Every grid line of that height has at most $2n/h$ points. Consequently, uniformly in p,*

$$\sum_{\ell \ni p} L_\ell^2 \le 16n^2(1+\log n), \tag{11}$$

$$\sum_{\ell \ni p} L_\ell^k \le 4(2n)^k\left(1+\frac{1}{k-2}\right) \qquad (k>2). \tag{12}$$

Proof. If v is primitive, the integer points on its line are precisely $p+\mathbb{Z}v$. Indeed, Bezout coefficients show that a real multiple of v with integer coordinates has an integer multiplier. A coordinate of absolute step h gives $L_\ell \le 1+(n-1)/h \le 2n/h$. For $h \ge 2$, counting the primitive points on the boundary of the square of radius h, with opposite signs identified, gives $4\varphi(h) \le 4h$; height one has four directions. Sum $4h(2n/h)^k$, using the harmonic bound when $k=2$ and an integral comparison when $k>2$. The same argument gives $\sum_{q\neq p} L_{pq} = \sum_{\ell\ni p}(L_\ell - 1)L_\ell \le \sum L_\ell^2$. □

Lemma 2.2 (Two uniform incidence kernels)**.** *For every fixed $\xi > 0$,*

$$\sum_{q \notin \ell(a,b)} L_{aq} L_{bq} \le C_\xi n^{2+\xi} \qquad (a \neq b), \tag{13}$$

$$\sum_{q \in \ell \cap V_n} L_{pq} \le C_\xi n^{1+\xi} \qquad (p \notin \ell). \tag{14}$$

Proof. The elementary divisor estimate $\tau(m) \le C_\epsilon m^\epsilon$ follows by writing $m = \prod l^{a_l}$. For sufficiently large primes, $a+1 \le 2^a \le l^{\epsilon a}$; the finitely many smaller primes have finite suprema of $(a+1)/l^{\epsilon a}$. Hence, with $\Delta_n = \max_{m \le 2n^2} \tau(m)$, $\Delta_n (1+\log n)^2 \le C_\xi n^\xi$.

For (13), write $q - a = eu$, $q - b = fv$, where $e, f > 0$ and u, v are primitive, and put $w = b - a$. Then $eu - fv = w$ and $f \mid \det(u, w)$. The exclusion of the baseline ensures $1 \le |\det(u,w)| \le 2n^2$, so for fixed u there are at most Δ_n choices of f. In dyadic height bands $H \le h(u) < 2H$, $J \le h(v) < 2J$, assume $H \le J$. There are $O(H^2)$ choices of u. The congruence $eu \equiv w \pmod f$ confines e to at most one residue class modulo f, by Bezout. A coordinate with $|u_i| \ge H$, together with $|v_i| < 2J$, confines e/f to an interval of length at most $4J/H$. Thus the number of candidates is $O(\Delta_n HJ)$. Their individual weights are at most $4n^2/(HJ)$. Interchange a, b when $H > J$, then sum the $O((1+\log n)^2)$ bands.

For (14), the cases of zero or one grid point are immediate. Otherwise write $\ell \cap \mathbb{Z}^2 = a + \mathbb{Z}v$ with primitive v, and let $D = |\det(v, p-a)| \in [1, 2n^2]$. For $q \in \ell$, put $g(q) = \gcd(|q_1 - p_1|, |q_2 - p_2|)$. Then $g(q) \mid D$. For each $e \mid D$, the condition $q \equiv p \pmod e$ restricts the line parameter to one residue class. Within $\|q-p\|_\infty \le R$, this gives at most $2R/(eh(v)) + 1$ choices. Since $g(q) \le \sum_{e \mid g(q)} e$ and relevant $e \le R$,

$$\sum_{\|q-p\|_\infty \le R} g(q) \le 3R\tau(D).$$

In the shell $R/2 < \|q-p\|_\infty \le R$, $L_{pq} \le 4ng(q)/R$, so the shell contributes at most $12n\tau(D)$. Sum dyadic shells and use the divisor estimate. □

2.1 Parity, chords and a uniform boundary estimate

Suppose F_r are constant on a fixed partition into R open rectangular cells, vanish on the common boundaries, and lie in $[0, M]$. Along a primitive line through a root of type r, the types alternate between r and $r + \bar{v}$.

Lemma 2.3 (Lattice-to-chord comparison)**.** *For every root and primitive direction of height h,*

$$\left| \sum_{q \in \ell(p,v) \setminus \{p\}} F_{\mathrm{typ}(q)}(q/n) - \frac{n}{2h} C_v (F_r + F_{r+\bar{v}})(p/n) \right| \le (4R+1)M. \tag{15}$$

Moreover $A_n = A + O_{R,T}(n^{-1})$.

Proof. Each parity subsequence has major-coordinate spacing $2h/n$. The number of points of any shifted one-dimensional lattice in an open interval differs from interval length divided by spacing by at most two. Sum over the two types and the R cells, then remove the root, costing at most M. A line coinciding with a common boundary has field identically zero on both sides of (15). For the mass statement, count each coordinate of each parity class in an open rectangle: the count is $n^2 \operatorname{area}(E)/4 + O(n)$. Sum the fixed number of cells and types. No assumption that n is even or a multiple of G is used. □

The regular grid in the concrete input permits a useful relative version of the mass estimate.

Lemma 2.4 (Mass on every integer grid)**.** *For nonnegative four-type fields constant on the open cells of a $G \times G$ grid and zero on its grid lines, let $A = \frac{1}{4}\sum_r \int F_r$ and $A_n = n^{-2}\sum_{p\in V_n} F_{\mathrm{typ}(p)}(p/n)$. For every integer $n \geq 2G$,*

$$A(1-2G/n)^2 \leq A_n \leq A(1+2G/n)^2. \tag{16}$$

Proof. For a coordinate parity $e \in \{0,1\}$, the number of integers of that parity in $(in/G, (i+1)n/G)$ is

$$N_{i,e} = \left\lceil \frac{(i+1)n - eG}{2G} \right\rceil - \left\lfloor \frac{in - eG}{2G} \right\rfloor - 1.$$

It lies between $n/(2G)-1$ and $n/(2G)+1$. Since the lower bound is nonnegative, the number of points of any specified parity type in a cell lies between the squares of these bounds. Multiply by that cell's nonnegative field value and sum over cells and types. The continuum contribution of a typed cell of value v is $v/(4G^2)$, giving (16). Boundary points have field zero, as required by the definition of the input. □

Lemma 2.5 (Uniform bounded variation)**.** *For a nonnegative field $F \leq M$ on R fixed rectangular cells, each of the four functions $u \mapsto c_{F,j}(x,u)^2$ satisfies*

$$\|c_{F,j}(x,\cdot)^2\|_\infty \leq M^2, \qquad \mathrm{Var}(c_{F,j}(x,\cdot)^2) \leq (4R+2)M^2,$$

uniformly in $x \in [0,1]^2$.

Proof. For $(1,u)$, the chord of a rectangle $(a,b)\times(c,d)$ through (x,y), for $u>0$, is

$$\left[\min(b, x+(d-y)/u) - \max(a, x+(c-y)/u)\right]_+ .$$

Its changes of branch occur only at slopes from the root to the four vertices. Commonly refine the at most $4R$ switch values. On each interval the total chord has the form $\alpha+\beta/u$, hence is monotone, as is its square since it is nonnegative. The square lies in $[0, M^2]$. At positive switches the values agree. At zero, a line may coincide with a common boundary; its possible endpoint jump costs at most one further M^2. The stated bound follows. Reflections and coordinate exchange give the other three families. □

This proof uses bounded variation, not a uniform derivative bound. For example, for a half-square and a root a distance ϵ above its boundary, a chord can have the form $\max(0, 1/2-\epsilon/u)$. Its derivative near $u = 2\epsilon$ is unbounded as $\epsilon \downarrow 0$, whereas its squared variation remains at most $1/4$.

2.2 A finite sieve for primitive directions

Let P be a fixed finite set of pairwise coprime odd integers greater than one; primality is unnecessary. Put

$$Q = \prod_{p\in P} p, \qquad D_P = \frac{3}{4}\prod_{p\in P}(1-p^{-2}), \qquad A_P = \prod_{p\in P}(1+p^{-2}).$$

For $e = (\epsilon,\eta) \in \{(0,1),(1,0),(1,1)\}$ and a nonnegative BV function w on $[0,1]$, define

$$K_{N,e}(w) = \sum_{h\leq N} \frac{1}{h^2} \sum_{\substack{1\leq a\leq h\,;\ (a,h)=1 \\ (h,a) \bmod 2 = e}} w(a/h).$$

Write $I = \int_0^1 w$, $M_w = \sup w$, and $V_w = \mathrm{Var}(w)$.

Lemma 2.6 (Finite-sieve upper bound)**.** *For every real $N \ge Q$,*

$$K_{N,e}(w) \le \frac{D_P}{3} I \log N + C_P(M_w + V_w), \qquad C_P = \left(5 + \frac{1}{4}\log Q\right) A_P. \tag{17}$$

The rational upper bound $C_P \le [5 + \frac{1}{4}\sum_{p\in P}(p-1)]A_P$ is also valid.

Proof. For real $x \ge 1$, omit the coprimality condition and denote the corresponding sum by $J_e(x; w)$. Counting alternating tags gives

$$\left| \sum_{\substack{1\le b\le k \\ b\equiv \eta \ (2)}} w(b/k) - \frac{k}{2} I \right| \le 2M_w + V_w.$$

To see this, place intervals of length $2/k$ to the left of the tags; the oscillation errors sum to at most V_w, and the omitted or cut end intervals have total length at most $4/k$. Also the parity harmonic sum differs from $\frac{1}{2}\log x$ by at most two, by comparing ordinary harmonic sums with integrals. The finite estimate $\sum_{k\le m} k^{-2} \le 2$ therefore gives

$$J_e(x; w) = \frac{I}{4}\log x + E_e(x), \qquad |E_e(x)| \le 5(M_w + V_w).$$

Nonnegativity of w allows the bound

$$\mathbf{1}_{(a,h)=1} \le \prod_{p\in P} (1 - \mathbf{1}_{p|a,\ p|h}).$$

Expand this finite product. For $A \subseteq P$, put $d_A = \prod_{p\in A} p$ and $s_A = (-1)^{|A|}$. Pairwise coprimality identifies simultaneous divisibility with divisibility by d_A; oddness preserves the parity class. Thus

$$K_{N,e}(w) \le \sum_{A\subseteq P} \frac{s_A}{d_A^2} J_e(N/d_A; w).$$

Since $N \ge Q$, every $N/d_A \ge 1$. Substituting the two-sided estimate for J_e and using $\sum s_A/d_A^2 = 4D_P/3$, $\sum 1/d_A^2 = A_P$, $\log d_A \le \log Q$, and $I \le M_w$ proves (17). It is essential to use a two-sided error estimate inside the signed inclusion-exclusion sum. Finally $\log p \le p - 1$ gives the rational constant. □

Only a one-sided bound is needed in this application. Consequently no infinite Mobius series, value of $\zeta(2)$, or assertion of probabilistic independence between parity and direction is required. The recorded choice of 175 factors gives $D_P < 76/125$; its product Q has 440 decimal digits. This large fixed threshold is compatible with the asymptotic quantifiers, but gives no useful small-grid claim. The factors and rational comparison are included in the companion data.

2.3 The common-deadline pair sum

For four fields F and a root p, define

$$\mathcal{B}_n(p; F) = \frac{1}{2n^2} \sum_{\substack{q,z\in V_n\setminus\{p\},\ q\ne z \\ p,q,z \text{ collinear}}} F_{\mathrm{typ}(q)}(q/n) F_{\mathrm{typ}(z)}(z/n). \tag{18}$$

The pair sum is ordered; the factor $1/2$ removes this duplication. A superscript Γ^c restricts it to nondesignated directions.

Proposition 2.7 (Uniform risk comparison)**.** *If $D_* \geq D_P$, then for $n \geq \max(2, Q)$,*

$$\mathcal{B}_n^{\Gamma^c}(p; F) \leq \mathcal{R}_{\mathrm{typ}(p)}(F; p/n) \log n + C_{\mathrm{geom}},$$

where, with $E_0 = (4R+1)M$, one may take

$$C_{\mathrm{geom}} = M^2 + 4ME_0 + 2E_0^2 + 6C_P(4R+3)M^2. \tag{19}$$

This constant is independent of n, p and of the field values within their fixed partition and bound.

Proof. For each direction let W_v be the weighted line sum with the root removed. The contribution is $(W_v^2 - \sum_q F_{\mathrm{typ}(q)}(q/n)^2)/(2n^2)$. Discard the nonnegative diagonal term. By (15), $W_v \leq nC_v(F_r + F_{r+\bar{v}})/(2h) + E_0$, while $C_v(F_r + F_{r+\bar{v}}) \leq 2M$. Expanding the square, and using at most $4h$ directions at height $h \leq n$, gives

$$\mathcal{B}_n^{\Gamma^c}(p; F) \leq \frac{1}{8} \sum_{\substack{v \text{ primitive}/\pm \\ h(v) \leq n}} \frac{C_v(F_r + F_{r+\bar{v}})(p/n)^2}{h(v)^2} + 4ME_0 + 2E_0^2.$$

Indeed, the cross term is at most $\sum_{h \leq n} 4h\, ME_0/(nh) \leq 4ME_0$; the squared error is at most $\sum_{h \leq n} 4h\, E_0^2/(2n^2) \leq 2E_0^2$. Represent nonaxis directions by the four families with slope a/h, $1 \leq a \leq h$. The diagonal overlap can be retained for an upper bound; the two axes contribute at most M^2 separately. Apply Lemma 2.6 to each of the twelve family/parity functions. Their supremum plus variation is at most $(4R+3)(2M)^2$, by Lemma 2.5, so the total sieve error, including the factor $1/8$, is at most $6C_P(4R+3)M^2$. For each nonzero parity v, the four integrals sum to $4g_{F_r + F_{r+v}}$. The coefficient is $\frac{1}{8} \cdot \frac{D_P}{3} \cdot 4 = D_P/6$. Increase it to $D_*/6$. The directions discarded at the start only increase the upper bound. □

For a general time-dependent input take $M = T$ once and for all. Any smaller bound on its terminal fields is also valid. The same constant works for every time, including every root's last-use time. For a fixed root with deadline τ, finite-product differentiation gives the exact identity

$$\frac{1}{n^2} \int_0^\tau \sum_{q,z}^{\Gamma^c} m_t(q) F_{\mathrm{typ}(z),t}(z/n)\, \mathrm{d}t = \mathcal{B}_n^{\Gamma^c}(p; F_\tau). \tag{20}$$

Indeed, differentiate each product $F_t(q)F_t(z)$, exchange q, z, and integrate. The index set is symmetric and consists of distinct points different from the root. Using separate deadlines for q and z would not give this identity.

2.4 Explicit constants for the 481-phase input

The input defined in Section 9.1 has $G = 256$. Every cumulative field is dominated by its terminal field, whose maximum is $6.25798212168 < 7$. Thus $R = 256^2 = 65536$ and $M = 7$ work at every time, although the sum of the phase durations is much larger than seven. Lemma 2.4 gives

$$A_n \geq A(1 - 512/n)^2 \qquad (n \geq 512).$$

For the 175 odd primes from 3 through 1049 used by the finite sieve, $\sum_{p \in P}(p-1) = 84164$. Also

$$S := \sum_{p \in P} p^{-2} < \sum_{k \geq 1} \frac{1}{4k(k+1)} = \frac{1}{4}.$$

The elementary symmetric sum of degree j in the numbers p^{-2} is at most S^j; expanding the finite product gives $A_P \le (1-S)^{-1} < 4/3$. Hence the rational choice $C_P = 84184/3$ is sufficient in Lemma 2.6. Substitution in (19), with $E_0 = 1835015$, gives

$$C_{\mathrm{geom}} = 8897332619623 < 10^{13}. \tag{21}$$

Consequently a continuum certificate $\mathcal{R}_{\mathrm{typ}(p)}(F_{\tau_p}; p/n) \le r_* = 9999/10000$ implies, uniformly over every used root,

$$\mathcal{B}_n^{\Gamma^c}(p; F_{\tau_p}) \le a \log n, \qquad a = \frac{9999}{10000} + \frac{1}{320000} = \frac{319969}{320000} < \frac{19999}{20000} = d, \tag{22}$$

provided $\log n \ge 3 \cdot 10^{18}$. This condition implies $n \ge Q$, since $\log Q \le 84164$, and it absorbs (21) with room to spare. It is only a sufficient threshold for the deterministic lattice comparison, not a claimed threshold for all stochastic error terms in the final theorem.

3 The stopped compensated process

Fix $n \ge 2$ and put $B = n^d$, $M_n = n^\gamma$, and $\epsilon_n = n^{-\zeta}$. A point is active, selected, geometrically dead, or positively retired. Initially every point is active, its weight is one, and $S = \varnothing$. Selection is permanent. Dead and retired points never return to the active set Q_t.

3.1 Line variables, rates and flows

Every designated line has occupancy $N_\ell = |S \cap \ell|$ and clock k_ℓ. For $N = 0, 1$ set

$$a_N(k) = \frac{2-N}{2-k}, \qquad b_0 = \frac{1}{2}, \quad b_1 = 1.$$

Every nondesignated line has a credit r_ℓ, initially one. It changes only while the line contains exactly one selected point. Before the global stopping time σ, define

$$\lambda_p(t) = \mathbf{1}_{p \in Q_t} \frac{m_t(p) w_p(t)}{n}, \qquad \Lambda_\ell = \sum_{p \in \ell} \lambda_p, \tag{23}$$

$$\dot{k}_\ell = \Lambda_\ell / a_{N_\ell}(k_\ell) \qquad (N_\ell = 0, 1), \tag{24}$$

$$\dot{r}_\ell = -r_\ell \Lambda_\ell \qquad (N_\ell = 1,\ \ell \text{ nondesignated}). \tag{25}$$

The other clock and line-credit derivatives are zero. For active p, put

$$H_p = \sum_{q \in Q_t \setminus \{p\}} \lambda_q \mathbf{1}_{\ell(p,q) \notin \Gamma,\ |S \cap \ell(p,q)| = 1}.$$

Here $\ell \notin \Gamma$ means that its direction is nondesignated. The weight flow is

$$\frac{\dot{w}_p}{w_p} = H_p + \lambda_p + \sum_{\substack{\ell \ni p \\ \ell \in \Gamma}} (\Lambda_\ell - \lambda_p) b_{N_\ell}. \tag{26}$$

The term H_p counts each actual cause q once, not once for each possible witness. All rates are set to zero after σ.

At a birth of q, use the pre-birth state to choose the branch, then perform these operations in order:

1. Select q and set its weight to zero.

2. Kill every active point that now lies on a line with two selected points, setting its weight to zero.
3. On each designated line having its first birth, halve the weights of the remaining active points.
4. Update occupancies and the activation flags of ordinary lines. The values of k_ℓ and r_ℓ are not reset.
5. Process any newly reached safety boundary and positive retirement.

For a fixed other point p, at most one line through p, q can cause a halving. Geometric death takes priority over positive saving.

3.2 Safety boundaries and the observation convention

A point retires positively at its first weight-cap hit $w_p = B$, at a hit $r_\ell = \rho$ on an ordinary line through it, or at its deadline τ_p. Its current positive weight is saved. Unused points retire at time zero with saved weight one.

Two additional variables define global guards. For distinct active a, b, let κ_{ab} be the sum of rates of birth causes $q \notin \ell(a, b)$ that would change both weights by a factor less than one. Set $\dot{\eta}_{ab} = -\kappa_{ab}\eta_{ab}$, initially $\eta_{ab} = 1$, and freeze this variable when either endpoint leaves the active state. For a designated line ℓ and an active point $q \notin \ell$, define the virtual deletion mass

$$\mathcal{V}_{\ell,q} = \sum_{y \in \ell \cap Q_t} \frac{w_y}{a_{N_\ell}(k_\ell)} \mathbf{1}_{\ell(q,y) \notin \Gamma,\ |S \cap \ell(q,y)| = 1}. \tag{27}$$

Globally stop at the first hit of $k_\ell = K$, $\eta_{ab} = 1 - \epsilon_n$, or $\mathcal{V}_{\ell,q} = M_n$, including an exceedance produced by a jump. Save the positive active credits and freeze the process. If no guard is hit, $\sigma = \infty$; the deterministic terminal time T is not itself a failure event.

When a mixed observation is saved at a root's positive boundary, take its value after geometric updates but before removing auxiliary roots because of simultaneous positive retirement. At a continuous hit take the boundary values before changing the flags. This convention prevents a tag witness from disappearing at the very instant it is observed. It does not alter the selected set or its birth rates. Auxiliary roots that retired strictly earlier contribute zero.

Proposition 3.1 (Construction and legality)**.** *For fixed n, the stopped process is well-defined on $[0, T]$, is adapted with right-continuous paths and left limits, and always produces a no-three-in-line set of at most $2n$ points.*

Proof. Give each point a Poisson candidate clock of rate B/n, together with independent uniform marks. At a candidate arrival, accept the birth if its mark is at most $m_t(p)w_p(t-)/B$ and the point is eligible in the pre-arrival state. The total candidate rate is nB, so there are finitely many arrivals on $[0, T]$ almost surely. Use the filtration revealing only arrivals and marks up to the current time, with the usual completion and right-continuous modification.

Between arrivals and mode changes, solve the finite-dimensional ODE. Within the stopped region the denominators are separated from zero: $k \le K < 2$, $r \ge \rho > 0$, and $\eta \ge 1 - \epsilon_n > 0$. Using w, rather than $\log w$, as a coordinate, each mode has a locally Lipschitz vector field. Extend it outside a compact neighborhood with a cutoff if necessary. The resulting flow is unique and depends continuously on its initial state. First hits of its closed guards are measurable: on any bounded time interval, the minimum distance of the continuous flow to a closed guard detects a hit. Composition at the preceding stopping time preserves adaptation.

Every positive retirement occurs at most once per point. Occupancies and line activation flags change only finitely often, deadlines and phases are finite, and a global stop occurs once. Resolve all simultaneous guards together and mark them as processed. Together with the finite candidate arrivals, this rules out an accumulation of updates. The construction gives the claimed path regularity.

Immediately before every birth the chosen point is active, hence is not on a line with two selected points. Adding it preserves legality; the subsequent deaths maintain this invariant. The row bound gives $|S| \le 2n$. □

For compensators, evaluate the intensity in the pre-jump state and use $\mathbf{1}_{t\le\sigma}$. The birth that triggers a global stop is counted. In time integrals one may instead use $\mathbf{1}_{t<\sigma}$, since the two differ on at most one time per path. Thinning gives, for a nonnegative predictable observation g,

$$\mathbb{E} \sum_{\text{births } (p,t)} g_p(t) = \mathbb{E} \int_0^T \sum_p \lambda_p(t) g_p(t)\, \mathrm{d}t. \tag{28}$$

For simple predictable functions this is independent increments and the uniform-mark rule; monotone approximation gives the general identity. Bounded signed observations follow by subtraction. All fixed-n rates and stopped observations used below are integrable.

4 Saved credit and the exact output identity

Define Z_p to equal w_p while p is active, zero after selection or geometric death, and its saved value after positive retirement or a global stop. Let C_p be the birth-eligible credit, including the global-stop indicator, and let V_p be saved credit. Then $Z_p = C_p + V_p$ and $0 \le Z_p \le B$.

Lemma 4.1 (One-point compensation)**.** *For each point p, Z_p is a bounded martingale and $\mathbb{E}Z_p(t) = 1$.*

Proof. While p is active, let $J_p(q) \in \{0, 1/2, 1\}$ be its multiplier at a birth of q. Selection or deletion gives zero, a first designated-line birth gives $1/2$, and otherwise it is one. The flow (26) is precisely

$$\dot{w}_p = w_p \sum_q \lambda_q (1 - J_p(q)).$$

By (28), the jump drift is its negative. Positive saving changes no value; subsequent drift is zero. A coincident geometric death has already been assigned multiplier zero. Thus the compensated drift vanishes, and boundedness makes the result a true martingale. Its initial value, including immediate saving of unused points, is one. □

Lemma 4.2 (Point law and birth intensities)**.** *The formula $\mathrm{d}\mathbb{P}^{(p)} = Z_p(T)\,\mathrm{d}\mathbb{P}$ defines a probability measure. At any bounded stopping time $\theta \le T$, $\mathbb{E}^{(p)}X = \mathbb{E}[Z_p(\theta)X]$ for nonnegative $\mathcal{F}_\theta$-measurable X. Before the root's saving boundary, the birth intensity of q under this law is $J_p(q)\lambda_q$.*

Proof. The martingale identity gives $\mathbb{E}[Z_p(T) \mid \mathcal{F}_\theta] = Z_p(\theta)$. One can obtain the stopping-time version from finite-valued dyadic approximations from above, right continuity and bounded convergence. For the intensity statement, enumerate the births, condition $Z_p(T)$ on each birth's post-jump sigma field, substitute $Z_p(t) = J_p(q)Z_p(t-)$, and then apply (28). Tonelli justifies the nonnegative birth sum. On time integrals $Z_p(t-) = Z_p(t)$ almost everywhere. □

The changed law forbids a birth that kills the root, but does not delete the self-rate term λ_p from the original deterministic weight ODE. Losing this distinction changes the compensation identity.

Proposition 4.3 (Exact remaining-exposure loss)**.** *With E_p and θ_p as in Section 1, (9) holds for every fixed n. Therefore*

$$\sup_p \mathbb{P}^{(p)}(E_p) \le \varepsilon \quad \Longrightarrow \quad \mathbb{E}|S_n| \ge (1-\varepsilon) n A_n.$$

Proof. Apply (28) to the birth count and use $\mathbb{E}Z_p(t) = 1$:

$$\mathbb{E}|S_n| = \frac{1}{n}\sum_p \int_0^T m_t(p)\mathbb{E}C_p(t)\,\mathrm{d}t = nA_n - \frac{1}{n}\sum_p \int_0^T m_t(p)\mathbb{E}V_p(t)\,\mathrm{d}t.$$

On a path with positive saving, $V_p(t) = Z_p(T)\mathbf{1}_{\theta_p \le t}$. Tonelli changes its integral to $Z_p(T)\int_{\theta_p}^T m_t(p)\,\mathrm{d}t$. Deadline saving has no remaining exposure. A path dying without positive saving has zero terminal credit and may be assigned $\theta_p = T$. The density identity proves (9). Finally the remaining exposure is between zero and the deterministic total $F_{\mathrm{typ}(p),T}(p/n)$; sum the probability bound. Events need not be independent or disjoint. □

Lemma 4.4 (Product formula and a positive lower bound)**.** *While p is active,*

$$w_p(t) = \exp\left\{\int_0^t H_p(u)\,\mathrm{d}u + \int_0^t \lambda_p(u)\left(1 - \sum_{\substack{\ell \ni p \\ \ell \in \Gamma}} b_{N_\ell}\right)\mathrm{d}u\right\} \prod_{\substack{\ell \ni p \\ \ell \in \Gamma}} a_{N_\ell}(k_\ell). \tag{29}$$

For large enough n, $w_p \ge 2^{-g-1}$ uniformly while active. The absolute value of the self-correction integral is at most $\max(1, g-1)TB/n = o(1)$.

Proof. Between births, $\mathrm{d}\log a_N(k)/\mathrm{d}t = \Lambda_\ell b_N$. At a first designated-line birth $a_1/a_0 = 1/2$, exactly matching the surviving point's halving. Subtract these factors from (26); all designated-line terms cancel except the displayed self-correction. Initially every factor is one. Since $H_p \ge 0$, $a_0 \ge 1$, $a_1 \ge 1/2$, and $\lambda_p \le B/n$, the asserted bounds follow. □

5 Mixed moments without independence

The estimates must retain leading coefficient one. Replacing all joint probabilities by an unspecified constant multiple of products would lose the narrow numerical margin. The following compensating factors separate ordinary-line dependence, designated-line dependence, and simultaneous changes caused from outside a pair's baseline.

Fix disjoint sets of labels C, D, called roots and witnesses, and keep their union $L = C \cup D$ fixed through the argument. If a witness q has been born, let b_q be a deterministic factor $e_q(t_q) \in [0,1]$ evaluated at its birth time; otherwise set it to zero. For a line put $m_\ell = |C \cap \ell|$. Define

$$\Phi_{C,D} = \mathbf{1}_{C \text{ active}} \prod_{q \in D} b_q \prod_{p \in C} w_p \prod_{\ell \notin \Gamma} r_\ell^{(m_\ell - 1)_+} \prod_{\ell \in \Gamma} a_{N_\ell}(k_\ell)^{-(m_\ell-1)_+}, \tag{30}$$

$$E_L = \prod_{\{a,b\} \subseteq L} \eta_{ab}, \qquad Y_{C,D} = E_L \Phi_{C,D}. \tag{31}$$

Only lines containing two roots contribute nontrivial factors, so the products are finite. The correction E_L uses the entire label set, not just the current roots. This makes it identical on the two sides of a witness-birth transition.

Lemma 5.1 (Generator and source bound)**.** *The positive source in the compensated drift of $Y_{C,D}$ is at most*

$$\sum_{q\in D}\beta_q(t)Y_{C\cup\{q\},D\setminus\{q\}}, \tag{32}$$

where $\beta_q = m_t(q)e_q(t)/n$ if no pair of labels in L shares a designated line. In general one may multiply each β_q by $\max(1,A_1)^g$, where $A_1 = 1/(2-K)$. Every other compensated drift and every auxiliary positive-retirement jump is nonpositive.

Proof. Fix a birth cause outside the current roots and group roots by the lines they share with it. On an active ordinary line, the m root compensations contribute $m\lambda_q$, while the factor r^{m-1} subtracts $(m-1)\lambda_q$; the group is killed at the birth. On a designated line of occupancy zero, the continuous contribution is $\lambda_q/2$, and the group multiplier is $(1/2)^m(a_0/a_1)^{m-1} = 1/2$. Occupancy one gives continuous contribution λ_q and multiplier zero. An empty ordinary line contributes nothing and has multiplier one.

Thus each group has a multiplier $J_i \in \{0, 1/2, 1\}$, and its combined relative drift is

$$\lambda_q\left[\sum_i(1-J_i)+\prod_i J_i - 1\right].$$

For $d_i = 1 - J_i \in [0,1]$, the finite Bonferroni inequality gives

$$0 \le \sum_i d_i - \left(1-\prod_i(1-d_i)\right) \le \sum_{i<j} d_i d_j.$$

One way to prove the upper bound is to introduce independent Bernoulli variables with success probabilities d_i just for this finite algebraic identity: if their sum is N, then $N - \mathbf{1}_{N\ge1} \le \binom{N}{2}$, and take expectations. This auxiliary proof does not assume independence in the geometric process. Different groups supply a pair of roots whose baseline does not contain q. The negative logarithmic drift of E_L, namely $-\sum_{\{a,b\}\subseteq L}\kappa_{ab}$, pays this upper bound. There is no jump cross term because E_L is continuous. If q is a root, its own continuous compensation and zeroing jump cancel; the shared-line correction cancels the other terms on its line. Thus internal causes introduce no extra positive drift.

It remains to consider an unborn witness $q \in D$. On a branch where all roots survive its birth, any shared ordinary line was previously empty, so its credit factor is one. A shared designated line containing $k \ge 1$ roots must previously have occupancy zero. Comparing the post-birth observation with the pre-birth source gives the exact factor

$$2^{-k}a_1^{-(k-1)}a_0^k = a_1.$$

The rate divided by the source's w_q is $m_t(q)/n$. The witness factor supplies $e_q(t)$, and E_L cancels exactly. There are at most g shared designated lines. This proves (32). Prior retirement of an auxiliary root only drops the observation to zero. □

Proposition 5.2 (Mixed moments at a common stopping time)**.** *Write $R = |L|$ and $F_q^\beta(t) = \int_0^t \beta_q(u)\,\mathrm{d}u$. On the stopped process,*

$$\mathbb{E}\Phi_{C,D}(t) \le (1-\epsilon_n)^{-\binom{R}{2}}\prod_{q\in D}F_q^\beta(t). \tag{33}$$

More precisely, for a deterministic $h \le T$ and any common bounded stopping time $\theta \le h$,

$$\mathbb{E}\Phi_{C,D}(\theta) \le (1-\epsilon_n)^{-\binom{R}{2}}\prod_{q\in D}F_q^\beta(h). \tag{34}$$

The entire fixed-label hierarchy is frozen at the same θ, using the postgeometry, prepositive observation convention.

Proof. For $A \subseteq D$, abbreviate

$$Y_A(t) = Y_{C\cup(D\setminus A),A}(t), \qquad h_q(t) = \int_t^h \beta_q(u)\,\mathrm{d}u.$$

Every Y_A uses the same label set $L = C \cup D$. Define the nonnegative backward polynomial

$$\mathcal{P}(t) = \sum_{A\subseteq D} Y_A(t) \prod_{q\in D\setminus A} h_q(t), \qquad 0 \le t \le h. \tag{35}$$

For $q \in A$, the source of Y_A in Lemma 5.1 is at most $\beta_q Y_{A\setminus\{q\}}$. In the summand indexed by $A \setminus \{q\}$, differentiating its factor h_q gives exactly the opposite term, since $h_q' = -\beta_q$. Thus all positive sources cancel term by term. The remaining drift and auxiliary-retirement jumps are nonpositive, so $\mathcal{P}$ is a supermartingale. For fixed n and fixed finite labels all these stopped observations are bounded, as are the deterministic coefficients; no unlocalized local-martingale expectation is being used.

At time zero all summands containing a witness vanish. The remaining root-only observation is at most one. Hence

$$\mathcal{P}(0) \le \prod_{q\in D} F_q^{\beta}(h), \qquad \mathbb{E}Y_{C,D}(\theta) \le \mathbb{E}\mathcal{P}(\theta) \le \mathcal{P}(0).$$

The first inequality follows from the $A = D$ term of the nonnegative sum. Finally divide by the pathwise lower bound $E_L \ge (1-\epsilon_n)^{\binom{R}{2}}$. This proves (34); set $h = \theta = t$ for (33). The integrals on the right always end at the deterministic h, not at the random θ.

For clarity, saving a boundary observation before simultaneous positive retirements is implemented by freezing the whole auxiliary hierarchy at θ. It does not restore an auxiliary root that retired earlier. Lemma 5.1 already holds after the geometric birth update without using the subsequent positive retirement, so it also applies to a birth that triggers this freeze. At a continuous boundary the values are unchanged by freezing. □

For fixed R, the first factor in (33) is $1 + o(1)$. Removing connection factors costs only a constant depending on R, g, K, ρ, because $r \ge \rho$ and $a_N \le 2/(2-K)$. In particular, products with j distinct born witnesses cost at most $C_R(t/n)^j$. A repeated root weight is bounded by $w^k \le B^{k-1}w$. A witness repeated many times is still just one birth and must not be charged as independent births.

For estimates under $\mathbb{P}^{(p)}$, put p among the roots of every member of the hierarchy, and freeze all of it at the first positive root boundary or global stop, capped at T. Use the postgeometry, prepositive observation convention. Lemma 4.2 then converts the expectation to the desired point-law moment. No witnesses born after this common boundary may be added. These conventions are needed for estimates at first crossings, not just at deterministic times. In the witness-free case the polynomial is the single observation itself. Its supermartingale property therefore also supplies conditional estimates between two observation times, as used below to remove ordinary-line credit.

6 Removing the global safety stops

All estimates here are proved for the stopped process. On it, $k \le K$, $r \ge \rho$, and $\eta \ge 1 - \epsilon_n$, so Proposition 5.2 is available before any probability bound on the global stop has been established.

6.1 Pair-correction and virtual-deletion guards

Lemma 6.1 (Pair-correction tail)**.** *For each fixed positive integer m,*

$$\mathbb{P}(\textit{pair-correction guard}) \le C_m n^{4-m+2m\xi+d(m-1)+m\zeta}. \tag{36}$$

Proof. For a fixed pair a, b, classify a cause changing both roots according to whether the two lines are ordinary or designated. Two ordinary lines require two distinct selected witnesses; one ordinary line requires one; two designated lines require only an intersection of two among finitely many directions.

In an m-fold expansion, regard distinct selected witnesses as vertices and the double-ordinary causes as edges of a multigraph. If there are j witnesses and c connected components, the first edge of a component costs $C_\xi n^{2+\xi}$ by (13). An edge adding one new witness costs $C_\xi n^{1+\xi}$ by (14). With both witnesses known, the cause is the intersection of two fixed lines and has at most one choice. Coincidence of those lines would put the cause on the excluded baseline. Thus the count is at most $C_m n^{j+(j-c)\xi} \le C_m n^{j+2m\xi}$. A component containing a designated/ordinary cause may be started with that cause, at cost $C_\xi n^{1+\xi}$. A double-designated cause has at most g^2 choices. This also covers mixed components and a witness appearing on different sides of different edges.

If the physical causes have $k \ge 1$ distinct labels, their rates supply n^{-m}, root repetitions cost $B^{m-k} \le B^{m-1}$, and the j distinct witnesses cost $C_m n^{-j}$ by Proposition 5.2. Root activity indicators can be replaced by constant multiples of root weights using Lemma 4.4. A label that is both an active cause and a selected witness gives zero. Therefore

$$\mathbb{E}\kappa_{ab}(t)^m \le C_m n^{-m+2m\xi} B^{m-1}.$$

Hitting the floor requires $\int_0^T \kappa_{ab}\,\mathrm{d}t \ge -\log(1-\epsilon_n) \ge \epsilon_n$. Apply Holder to the time integral, then Markov and a union bound over at most n^4 pairs. This is (36). □

Lemma 6.2 (Virtual-deletion tail)**.** *For each fixed positive integer m,*

$$\mathbb{P}(\textit{virtual-deletion guard}) \le C_m \sum_{a=1}^{m} n^{3-\gamma m+d(m-a)+a\xi}. \tag{37}$$

Proof. Fix a designated ℓ and an active $q \notin \ell$. In an m-fold expansion of (27), suppose there are a distinct leaves on ℓ. The lines from q to these leaves are distinct, and their selected witnesses are distinct: a common witness on two such lines would be q, which is active. By (14) the number of leaf/witness choices is at most $C_m n^{a(1+\xi)}$. Source prices supply n^{-a}, and repeated leaves cost B^{m-a}. The normalized weight is at most $2w$. Consequently

$$\mathbb{E}[\mathbf{1}_{q \text{ active}} \mathcal{V}_{\ell,q}^m] \le C_m \sum_{a=1}^{m} B^{m-a} n^{a\xi}.$$

Apply the same fixed-label hierarchy at the first crossing; no union over observation times is necessary. Specifically, include q and all distinct leaves as roots, the selected labels as witnesses, and apply (34) with deterministic terminal time T. The activity indicator of q is bounded by a constant times its positive weight by Lemma 4.4. There are $O(n)$ designated lines and n^2 possible external points. Markov at level n^γ gives (37). □

The slope in m in (36) is $-(1-d-2\xi-\zeta) < 0$. Since $\xi < d$, the largest exponent in (37) occurs at $a = 1$ and equals $3 - d + \xi - m(\gamma - d)$. Both probabilities are therefore smaller than any prescribed fixed power of n^{-1}, after choosing one sufficiently large fixed moment order. The process itself is unchanged by that choice of proof order.

For the concrete parameters $d = 19999/20000$, $\gamma = 39999/40000$, and $\xi = \zeta = 10^{-6}$, one may already take $m = 2^{18} = 262144$. Including the factor B for a change to any point law, the two exponents are respectively

$$4 - m + 2m\xi + d(m-1) + m\zeta + d = -8.320768, \qquad 3 - \gamma m + d(m-1) + \xi + d = -3.553599.$$

Only their negativity is needed. The constants depending on this fixed order need not be evaluated to obtain an asymptotic bound.

6.2 The protected law of a designated line

Fix a designated line and one of its occupancy stages $N = 0, 1$. Starting at the stage's entry time U, suppress its next birth through the density process

$$D_t = \mathbf{1}_{\text{no new line birth in } (U,t]} \exp\left(\int_U^t \Lambda_\ell(s)\,\mathrm{d}s\right),$$

stopped and saved at the relevant guards or T. Before U its value is one. The birth at U, if any, starts the stage and is not counted again as its next birth. The continuous drift $D\Lambda_\ell$ cancels the zeroing-jump drift. Also

$$\int_U^t \Lambda_\ell\,\mathrm{d}s = \int_{k(U)}^{k(t)} a_N(k)\,\mathrm{d}k \le 2\log\frac{2}{2-K}.$$

Thus D is a bounded martingale and defines a probability law conditionally on the entry history. Its upper bound is the fixed constant A_0^2, where $A_0 = 2/(2-K)$. On a crossing occurring before the next line birth, $D \ge 1$, so the original crossing probability is at most its probability under the protected law. In particular this comparison does not introduce the upper-density factor A_0^2. Under the protected law the next line birth has rate zero; the external birth rates remain their original rates.

Normalize each active line weight by $c_p = w_p/a_N(k_\ell)$, and save this *normalized* value at positive retirement. It is not subsequently divided by a changing a_N. For a fixed cell/type subset of the line, let M be the sum of current and saved c_p. Define its entropy sum using

$$h(c) = c\left(1 + \log\frac{2B}{c}\right), \quad h(0) = 0, \qquad H = \sum h(c_p),$$

again with saved terms frozen. Under the protected law the surviving positive drift of each credit is $g_p = \lambda_p(1 - b_N)c_p$. This follows by subtracting $\mathrm{d}\log a_N/\mathrm{d}t = \Lambda_\ell b_N$ from the original ODE and deleting only the line-birth jump rates. It is $\lambda_p c_p/2$ at occupancy zero and zero at occupancy one.

Lemma 6.3 (Credit, entropy and whole-event energy)**.** *Stop further at $M \le C_0 n$. There are compensated martingales Y_M, Y_H and a nonnegative cumulative cost V such that*

$$M = M(U) + D_M + Y_M, \qquad D_M = O(B), \tag{38}$$
$$H = H(U) + D_H - V + Y_H, \qquad D_H = O(B\log n), \tag{39}$$

and, before the virtual guard,

$$\langle Y_M\rangle \le Cn^\gamma V, \qquad |\Delta Y_M| \le Cn^\gamma, \tag{40}$$
$$\langle Y_H\rangle \le Cn^\gamma(\log n)^2 V, \qquad |\Delta Y_H| \le Cn^\gamma \log n. \tag{41}$$

The constants are independent of the admissible entry history.

Proof. For a multiplier $J \in \{0, 1/2, 1\}$, direct algebra gives

$$h'(c)c(1-J) + h(Jc) - h(c) = -c(J\log J - J + 1).$$

The cost is c for deletion, $c(1-\log 2)/2$ for halving, and zero otherwise. It is comparable by fixed positive constants to the credit decrease $c(1-J)$. Define V to be the rate integral of the *sum of these costs for each complete birth event*. Compensating gives (38) and (39). Since $\lambda_p \le B/n$ and the credit sum is capped by $C_0 n$, the first drift is $O(B)$. The active credits have the positive lower bound supplied by Lemma 4.4 divided by A_0, so $h'(c) = \log(2B/c) = O(\log n)$ on them. This gives the entropy drift estimate.

In one external birth event, the ordinary-line deletion of normalized mass is at most M_n, and effects through other designated directions touch at most g points, each of credit at most $2B$. Its total credit decrease Δ is therefore at most $M_n + 2gB = O(n^\gamma)$. The square of the *whole* decrease satisfies $\Delta^2 \le Cn^\gamma \Delta$. The absolute entropy decrease is at most $C\log n$ times this credit decrease, including deletion to zero. Comparing the decrease to the event's total cost and integrating its birth rate proves (40)–(41). These estimates retain all cross terms of simultaneous changes. Summing the squares of individual point changes would not prove the required bracket bound. The birth that causes a global stop is included: its decrease is controlled by the virtual-guard bound in the state immediately before that birth. Positive saving itself changes neither the credit ledger nor its entropy. □

Lemma 6.4 (Exponential crossing estimate)**.** *Let Y be a compensated birth martingale with $Y_0 = 0$, jumps bounded in absolute value by b, and continuous predictable quadratic variation Q. For $x, v > 0$,*

$$\mathbb{P}(\exists t \le T : Y_t \ge x,\ Q_t \le v) \le \exp\left(-\frac{x^2}{2(v + bx/3)}\right). \tag{42}$$

Proof. For $0 < \theta b < 3$ and $|u| \le b$, the Taylor series and $k! \ge 2 \cdot 3^{k-2}$ for $k \ge 2$ give

$$e^{\theta u} - 1 - \theta u \le \psi u^2, \qquad \psi = \frac{\theta^2}{2(1-\theta b/3)}.$$

Stop at the first $Y \ge x$, capped at T. The stopped exponential $\exp(\theta Y - \psi Q)$ is bounded by $\exp(\theta(x+b))$. Its compensated drift is nonpositive by the displayed inequality, so its expectation is at most one. Since Q is continuous and nondecreasing, a crossing with $Q \le v$ exists precisely when the first crossing has $Q \le v$. On this event the terminal stopped exponential is at least $\exp(\theta x - \psi v)$. Markov gives $\exp(-\theta x + \psi v)$; take $\theta = x/(v + bx/3)$. This proof uses a bounded stopped exponential, rather than assuming that an unlocalized exponential local martingale has mean one. □

Proposition 6.5 (Clock stops are rare)**.** *For each fixed positive tolerance and admissible fixed data, the failure probability of the simultaneous designated-line credit bound is at most*

$$Cn\exp\left(-c\frac{n^{1-\gamma}}{\log n}\right).$$

In particular, the probability of a designated clock reaching K is smaller than every fixed inverse power of n.

Proof. Stop the cumulative cost at $C_1 n\log n$. Since $H \ge 0$, $H(U) = O(n\log n)$, and $D_H = o(n\log n)$, reaching this cost cap, for sufficiently large C_1, forces an upward fluctuation of Y_H of order $n\log n$. Apply Lemma 6.4 with $v = O(n^{1+\gamma}(\log n)^3)$ and $b = O(n^\gamma \log n)$. An

increase of M by a fixed multiple of n similarly forces a fluctuation of Y_M of order n, with $v = O(n^{1+\gamma}\log n)$ and $b = O(n^{\gamma})$. Both exponents have order $n^{1-\gamma}/\log n$.

At the first birth on the line, $(w/2)/a_1 = w/a_0$; the selected point is removed. The saved ledger is unchanged and the moving normalized credit cannot increase. At the second birth, all remaining moving credit disappears. To treat the occupancy-one stage, condition on its actual entry history and the preceding good credit bound. At time zero the ledger of a cell/type piece equals its lattice-point count $L_E \le n$. On a good occupancy-zero history it is at most $L_E + \varepsilon n$, and the first line birth cannot increase it. The lower and upper credit bounds from Lemma 4.4 give $h(c) \le C(\log n)c$, including saved positive credits. Consequently both initial credit and entropy bounds hold uniformly at the occupancy-one entry time. Apply the same conditional bound to every such history and average it. The union is only over the two stages, not over possible histories or starting times.

There are $O(n)$ designated lines and a fixed number of cell/type pieces. The estimates control whole time intervals, so no separate union over phases is needed. Summing their crossing probabilities preserves the displayed bound. On the resulting good event, integrate the moving normalized credit with the masks to obtain

$$k_\ell \le \text{deterministic line capacity} + C_{\text{data}} T\varepsilon + O(n^{-1}).$$

Choose $\varepsilon > 0$ first so that the second term is smaller than half of κ_H, then choose n large. The capacity condition (5) precludes hitting K. Positive retirement need not be rare for this argument: its credit was retained in the ledger throughout. □

6.3 Explicit clock constants for the verified input

For the input used in the application,

$$g = 16, \quad K = 255/128, \quad A_0 = 256, \quad G = 256, \quad T = \frac{30088384743729}{25000000000} < 2^{11}.$$

Write $C_{\text{line}} = 4G = 1024$. The verified designated-line capacity bound has margin

$$\kappa_H = \frac{69405753}{80000000000}, \qquad \varepsilon_H = \frac{\kappa_H}{8TC_{\text{line}}} = \frac{115676255}{1314581588376682496} > 2^{-34}.$$

The following explicit estimate is not needed to compute a final threshold for all five stopping causes. It records how the history-uniform clock argument applies to this particular input.

Corollary 6.6 (A quantitative clock budget)**.** *For all sufficiently large n,*

$$\sup_p \mathbb{P}^{(p)}(\textit{clock is an early cause}) \le 221184\, n^{1+d} \exp\left(-\frac{\varepsilon_H^2 n^{1-\gamma}}{2^{18}\log n}\right). \tag{43}$$

In fact, with $d = 19999/20000$ and $\gamma = 39999/40000$,

$$\log n \ge 10^7 \quad \Longrightarrow \quad \sup_p \mathbb{P}^{(p)}(\textit{clock is an early cause}) < 2\cdot 10^{-6}. \tag{44}$$

Proof. Assume first $n^{1-d} \ge 30T$. The self-correction in (29) has absolute value at most $15TB/n \le 1/2$, so $w_p \ge 2^{-17}$ while active. Every positive normalized credit, including a saved credit, therefore satisfies

$$2^{-25} \le c_p \le 2B.$$

For $L_n = 32\log n$ and $n \ge 2$, this implies $h(c) \le L_n c$ and $0 \le h'(c) \le L_n$ on positive active credits. Stop temporarily while $M \le 3n$. The drift terms in Lemma 6.3 obey

$$0 \le D_M \le \tfrac{3}{2}TB, \qquad 0 \le D_H \le L_n D_M.$$

For a complete external birth let v be its entropy cost and Δ its credit loss. Since $\log 2 < 3/4$, deletion and halving both satisfy $\Delta/4 \le v \le \Delta$. The virtual guard and at most 16 other designated-line intersections give $\Delta \le n^\gamma + 32B \le 33n^\gamma$. Hence

$$\langle Y_M \rangle \le 132 n^\gamma V, \qquad |\Delta Y_M| \le 33 n^\gamma,$$
$$\langle Y_H \rangle \le 132 n^\gamma L_n^2 V, \qquad |\Delta Y_H| \le 33 n^\gamma L_n.$$

The bound on an entropy jump follows directly from $h(c) - h(Jc) \le L_n c(1-J)$, including $J = 0$; it does not assume a bounded derivative at zero.

Consider a stage starting with $M(U) \le 2n$, and any $0 < \varepsilon \le 1$ for which $n^{1-d} \ge 3T/\varepsilon$. Stop at the first of a credit increase εn, cost $V = 4L_n n$, an original guard, and T. Credit increases and V are continuous, so before this stop $M \le 3n$, and $D_M \le \varepsilon n/2$. If the cost boundary is reached first, $H \ge 0$, $H(U) \le 2L_n n$ and $D_H \le L_n \varepsilon n/2$ force $Y_H \ge L_n n$. Lemma 6.4, with $v = 528 n^{1+\gamma} L_n^3$ and $b = 33 n^\gamma L_n$, bounds its probability by $\exp[-n^{1-\gamma}/(1078 L_n)]$. If the credit boundary is reached first, $Y_M \ge \varepsilon n/2$ and $\langle Y_M \rangle \le 528 L_n n^{1+\gamma}$, giving $\exp[-\varepsilon^2 n^{1-\gamma}/(4268 L_n)]$. As $4268 \cdot 32 < 2^{18}$, their sum is at most

$$2 \exp\left(-\frac{\varepsilon^2 n^{1-\gamma}}{2^{18} \log n} \right).$$

This bound is conditional on the entry history and its constants are independent of that history.

At the first line birth the ledger cannot increase. Applying the last estimate first at occupancy zero and then conditionally at occupancy one shows that $M_E(t) \le L_E + 2\varepsilon n$ fails with at most twice the displayed probability. The 16 primitive unoriented directions of height at most three satisfy $\sum_{v \in \Gamma} (|v_1| + |v_2|) = 54$, so they determine at most $54n$ lines. Each line has at most 1024 nonempty cell/type pieces, by Lemma 8.1 and the corresponding axis count. Thus the simultaneous ledger bound fails under the original law with probability at most

$$221184\, n \exp\left(-\frac{\varepsilon^2 n^{1-\gamma}}{2^{18} \log n} \right).$$

On its complement, integration against the masks gives

$$k_\ell(t) \le \frac{1}{n} \sum_{q \in \ell} F_{\mathrm{typ}(q),T}(q/n) + 2\varepsilon T C_{\mathrm{line}}.$$

For this input the field bound is 7 and there are 256^2 common rectangles. The lattice-to-chord argument without removing the root has error at most E_{cap}/n, where $E_{\mathrm{cap}} = 4 \cdot 256^2 \cdot 7 = 1835008$. Take $\varepsilon = \varepsilon_H$ and $n \ge 4E_{\mathrm{cap}}/\kappa_H$. The lattice error and ledger error are each at most $\kappa_H/4$, so $k_\ell < K$. Finally $Z_p(T) \le B$ changes to any point law at cost at most $B = n^d$, proving (43).

To verify the stated threshold, put $s = \log n \ge 10^7$. The rational bounds $\varepsilon_H > 2^{-34}$, $T < 2^{11}$, $\kappa_H > 2^{-11}$, and $E_{\mathrm{cap}} < 2^{21}$ imply all three preliminary conditions, since $n^{1-d} = e^{s/20000} \ge e^{500} > 2^{500}$. The right side of (43) is at most

$$2^{18} \exp\left(2s - \frac{e^{s/40000}}{2^{86} s} \right).$$

The function $e^{s/40000}/s^2$ increases for $s \ge 80000$. At $s = 10^7$, use $e^{250} > 2^{250}$ and $10^{14} < 2^{47}$ to obtain $e^{s/40000}/(2^{86} s) > 4s$; this persists for larger s. The result is bounded by $2^{18} e^{-2s} < 2 \cdot 10^{-6}$. This is only a threshold for the clock cause, not a value of the eventual threshold in the final existence theorem. □

Combining the three guards shows, for every fixed $A_0' > 0$,

$$\mathbb{P}(\sigma \le T) \le C_{A_0'} n^{-A_0'}. \tag{45}$$

Under any point law this is at most B times as large, hence is $o(1)$ uniformly in the root. This establishes global-stop smallness before the weight-cap and ordinary-line tag estimates below.

7 Ordinary-line risk under the point law

Fix a root p and its deterministic deadline $\tau = \tau_p$. All observations below are stopped and saved at the root's first positive boundary or global stop. A quantity explicitly restricted to the active root is zero afterwards. Write $F_q(t) = \int_0^t m_u(q)\,\mathrm{d}u$, and let

$$I_p = \int_0^{\theta_p \wedge T} \mathbf{1}_{p \text{ active}} H_p(t)\,\mathrm{d}t$$

be its integrated ordinary-line hazard. Earlier root death has $\mathbb{P}^{(p)}$-probability zero. We will bound I_p by a proxy having sharp fixed-order moments, plus an error small in L^1. The distinction between those two kinds of estimate is essential.

7.1 Removing ordinary-line credit at additive cost

For an ordinary line through p, let $H_\ell(t) = r_\ell(t) \sum_{q \in \ell \setminus \{p\},\ q \text{ active}} w_q(t)$. The two-root, no-witness case of Proposition 5.2, followed by the conditional density identity, gives

$$\mathbb{E}^{(p)}[\mathbf{1}_{p \text{ before boundary at } t} H_\ell(t) \mid \mathcal{F}_u] \le (1-\epsilon_n)^{-1} \mathbf{1}_{p \text{ before boundary at } u} H_\ell(u) \quad (u \le t). \tag{46}$$

For each leaf the witness-free observation $\eta_{pq} w_p r_\ell w_q$ is a supermartingale. Under the point law this bounds the conditional expectation of $\eta_{pq} r_\ell w_q$; the ratio of its correction factors is at most $(1-\epsilon_n)^{-1}$. Conditional division is performed only where $Z_p(u) > 0$; its complement is null under the point law. The auxiliary-root sum excludes p, so no self-square is silently included in this two-root argument.

One selected witness $z \in \ell$ and two leaves give

$$\mathbb{E}^{(p)}[\mathbf{1}_{p \text{ before boundary},\ z \in S_t} H_\ell(t)^2] \le C L_\ell (L_\ell + B) t/n. \tag{47}$$

For distinct leaves the connection factor is exactly r_ℓ^2; a repeated leaf is paid using $r^2 w^2 \le Brw$. The one-leaf version is at most $CL_\ell t/n$. All labels here lie on the same ordinary line, so there is no designated-line source multiplier.

Since r starts moving at the first line birth, $1 - r_\ell(t) = \int_{\text{activation}}^t r_\ell \Lambda_\ell\,\mathrm{d}u$. The correct self-rate bound is

$$r_\ell \Lambda_\ell \le H_\ell/n + B/n. \tag{48}$$

Multiplying the leaf sums in the hazard by r_ℓ therefore loses a nonnegative amount whose expectation is at most

$$C n^{-3} \sum_{\ell \ni p} (L_\ell^3 + B L_\ell^2) + CBn^{-3} \sum_{\ell \ni p} L_\ell^2 = O(1 + (B/n)\log n) = o(\log n). \tag{49}$$

For completeness, insert the integral for $1 - r$, use (46) to move the future H to the earlier time, cover activation by $\sum_{z \in \ell} \mathbf{1}_{z \in S_u}$, and apply (47). The second term in (49) is the separate self-rate contribution from (48). The line sums are Lemma 2.1. This is an additive error, not a constant-factor loss in the leading $\log n$ coefficient.

7.2 Long lines: replace the leaf sum, retain the birth time

Call a line long when $L_\ell \geq B \log n$. Fix a selected witness $z \in \ell$, and put

$$H = r_\ell \sum_{q \in \ell \setminus \{p\},\ q \text{ active}} m_t(q) w_q, \qquad D_z(t) = \sum_{q \in \ell \setminus \{p,z\}} m_t(q).$$

For every fixed integer k, the mixed hierarchy gives

$$\mathbb{E}^{(p)}[\mathbf{1}_{p \text{ before boundary},\ z \in S_t} H^k] \leq (1 + o(1)) \frac{F_z(t)}{n} [D_z(t)^k + C_k B L_\ell^{k-1}]. \tag{50}$$

Indeed, the all-distinct leaves have exactly the required connection r_ℓ^k and unit source price. A term with $a < k$ distinct leaves costs $B^{k-a} L_\ell^a \leq B L_\ell^{k-1}$, since $B \leq L_\ell$. There are finitely many repetition patterns for fixed k. No division by a positive lower bound on F_z is used; if it is zero the corresponding birth event is null.

For fixed $\epsilon > 0$, use $H \mathbf{1}_{H > D + \epsilon L} \leq H^k / (D + \epsilon L)^{k-1}$. Since $D \leq L$, the main term is at most $L(1+\epsilon)^{-(k-1)}$, while the repeated-leaf term is $C_{k,\epsilon} B$. Sum over witnesses, long lines and time, with the hazard factor $1/n$. As $B/L \leq 1/\log n$, the positive error R_{long} made by replacing the leaf sum by $D_z(t)$ satisfies

$$\limsup_{n \to \infty} \sup_p \frac{\mathbb{E}^{(p)} R_{\text{long}}}{\log n} \leq C\epsilon + C(1+\epsilon)^{-(k-1)}. \tag{51}$$

First choose ϵ, next a fixed sufficiently large k, and only then let n grow. This proves an $o(\log n)$ L^1 error for the same process.

For a witness born at time u, define its deterministic remaining cost and birth price by

$$A_z(u) = \frac{1}{n} \int_u^\tau D_z(t)\, \mathrm{d}t \leq \frac{T L_{pz}}{n}, \qquad b_z = \frac{1}{n} \int_0^\tau m_u(z) A_z(u)\, \mathrm{d}u.$$

Let P_{long} be the sum of A_z over long-line witnesses born before the root boundary, and let $M_{\text{long}} = \sum_z b_z$. Insert $A_z(u)/(T L_{pz}/n) \leq 1$ as a deterministic birth-retention factor in the mixed hierarchy, and use (34) with deterministic horizon $h = \tau$. The common stopping time limits which witnesses are included; it does not replace τ inside the deterministic cost or birth-price integral. If $T = 0$ there are no witnesses and the assertion is immediate. It follows that for every fixed m,

$$\|P_{\text{long}}\|_{L^m(\mathbb{P}^{(p)})} \leq M_{\text{long}} + o(\log n). \tag{52}$$

Here are the non-generic terms in this assertion. A repeated witness block of size $r \geq 2$ has total price at most $C_r n^{-(r+1)} \sum_\ell L_\ell^{r+1} = O_r(1)$, whereas a singleton block costs $O(\log n)$. Thus repetitions lose a power of $\log n$. For two distinct witnesses sharing a designated line, that line cannot pass through the root. By (14), the weighted configuration count is

$$\sum_{z_1} L_{pz_1} \sum_{\substack{z_2 \text{ on a designated} \\ \text{line through } z_1}} L_{pz_2} \leq C n^{3+\xi} \log n.$$

The two birth-and-cost factors supply n^{-4}, giving $O(n^{-1+\xi} \log n)$. Remaining singleton factors cost $O(\log n)$ each. A finite union over pairs covers multiple exceptions. The all-distinct, unshared case keeps the exact product of the b_z, with leading factor $1 + o(1)$.

7.3 Short stars and their exceptional configurations

An active short line through p has one selected witness z. Each active leaf $q \neq p$ on that line gives a star (p, q, z). Let

$$X_t = \sum_{\text{short stars}} m_t(q) r_{\ell(p,q)} w_q, \qquad D_t = \frac{1}{n} \sum_{\text{deterministic short stars}} m_t(q) F_z(t).$$

The actual sum is zero after the root boundary. In the deterministic sum, p, q, z are distinct and collinear on a short ordinary line; there is no selected-point condition. Put $t_0 = n^{\gamma-1}$.

Lemma 7.1 (Short-star moment estimate)**.** *For each fixed positive integer m, uniformly for $t_0 \le t \le T$ and all roots,*

$$\mathbb{E}^{(p)} X_t^m \le (1 + o(1)) D_t^m + \epsilon_m(n)(tn \log n)^m, \qquad \epsilon_m(n) \longrightarrow 0. \tag{53}$$

More explicitly, for all sufficiently large n (depending on the fixed m), one may use

$$\mathbb{E}^{(p)} X_t^m \le [1 + C_m n^{-\zeta}] D_t^m + C_m (n^{d-\gamma} + n^{\gamma-1} + n^{-1+\xi})(tn \log n)^m. \tag{54}$$

Proof. In an m-fold product let a be the number of distinct leaves and j the number of distinct witnesses. The active root makes the selected witness on each of its lines unique, so $1 \le j \le a \le m$. A leaf coinciding with a witness or the root gives zero. If $j < m$, the first leaf and witness of each star line have $O(n^2 \log n)$ choices by (11). Additional leaves on that line have at most $B \log n$ choices, and repeated weights cost B^{m-a}. The source price is $C_m (t/n)^j$. Dividing this bound by $(tn \log n)^m$ gives

$$C_m \left(\frac{B}{nt}\right)^{m-j} (\log n)^{a-m} = o(1),$$

uniformly since $B/(nt) \le n^{d-\gamma}$.

It remains to treat $j = a = m$, with distinct star lines. The following list exhausts the extra connections among different stars.

(i) Two labels share a designated line. Fix the first label. The line is outside the root, so (14) bounds the second label together with its star partner by $Cn^{1+\xi}$, instead of the generic $O(n^2 \log n)$ row. This covers leaf–leaf, leaf–witness and witness–witness sharing.

(ii) A line connecting two leaves has length greater than n^γ. For a fixed first leaf there are $O(n^{2-\gamma})$ possible second leaves: sum the height counts up to $O(n^{1-\gamma})$. Its star witness has at most $B \log n$ choices. Relative to a generic star row the saving is $O(n^{d-\gamma})$.

(iii) A leaf–leaf line of length at most n^γ is active through a new witness. There are at most n^γ positions for it and an additional source price $C_m t/n$, giving $O(n^{\gamma-1})$.

(iv) That line is active through an existing third star witness. It cannot be either leaf's own witness, since then their star lines would coincide. For a third witness use $\sum_{z \in \ell(q_i, q_j)} L_{pz} \le Cn^{1+\xi}$, keeping all m original birth prices. This again saves $n^{-1+\xi}$. An active leaf or the root cannot itself be a selected witness.

Outside these cases the additional ordinary-line credits are all one, there are no designated-line connections, and the star credits exactly match the mixed-moment factors. The source price is the exact product of $F_z(t)/n$; keeping the masks produces D_t^m. Each single exception is bounded after dropping other restrictions, so a finite union over label pairs also covers overlapping exceptions. Removing remaining factors costs only C_m, including $\rho^{-O(m^2)}$. The savings above and the correction $(1 - \epsilon_n)^{-O(m^2)} = 1 + O_m(n^{-\zeta})$ prove (54) and hence the assertion. □

For the parameters used here, $d-\gamma=\gamma-1=-1/40000$ and $-1+\xi=-999999/1000000$, while $\zeta=10^{-6}>0$. All these errors vanish for every fixed order. No uniform control of C_m as m grows is asserted or used.

Minkowski's inequality now yields

$$\left\|\int_{t_0}^{\tau} X_t/n\,\mathrm{d}t\right\|_m \le \int_{t_0}^{\tau} D_t/n\,\mathrm{d}t + o(\log n), \tag{55}$$

with the integral zero when $\tau<t_0$. For earlier times the one-witness, one-leaf bound gives $\mathbb{E}^{(p)}X_t=O(tn\log n)$. Its integrated contribution is $O(t_0^2\log n)=o(\log n)$ in L^1. No higher moment for this early-time error is asserted.

7.4 Hazard concentration and the weight cap

Collect the credit-removal error, long-line leaf replacement error, and early short-star contribution into $R_n\ge 0$, so $\sup_p \mathbb{E}^{(p)}R_n=o(\log n)$. Define

$$Y_n = P_{\mathrm{long}} + \int_{t_0}^{\tau} X_t/n\,\mathrm{d}t.$$

Then $I_p\le Y_n+R_n$, and (52) and (55) give, for every fixed m,

$$\|Y_n\|_m \le M_{\mathrm{long}} + M_{\mathrm{short}} + o(\log n) \le \mathcal{B}_n^{\Gamma^c}(p;F_\tau) + o(\log n).$$

The last inequality is (20): the division into long and short lines is symmetric in leaf and witness, and all four fields use the same τ. Proposition 2.7 and (7) therefore give the following summary, uniformly in the root and for every fixed positive integer m:

$$I_p \le Y_n + R_n, \qquad \|Y_n\|_{L^m(\mathbb{P}^{(p)})} \le (r_* + o_m(1))\log n, \qquad \mathbb{E}^{(p)}R_n = o(\log n). \tag{56}$$

For any $b>r_*$, choose $\epsilon>0$ with $r_*+\epsilon<b-\epsilon$. Markov on R_n, followed by the fixed-m moment bound, shows

$$\limsup_{n\to\infty}\sup_p \mathbb{P}^{(p)}(I_p > b\log n) \le \left(\frac{r_*+\epsilon}{b-\epsilon}\right)^m.$$

The right side tends to zero as the fixed proof order m is chosen larger. This quantifier order does not require constants uniform over all m, or an order depending on n. By (29), hitting $B=n^d$ requires $I_p\ge d\log n - O(1)$. Choose $r_*<b<d$. We have proved

$$\sup_p \mathbb{P}^{(p)}(\text{weight cap is an early saving cause}) = o(1). \tag{57}$$

8 Ordinary-line tags and the final loss estimate

The ordinary-line tag $r_\ell=\rho$ must be controlled uniformly over all directions. We use fixed moment orders throughout this section. Retaining the line's height and counting only the spatial cells that it meets avoids a growing moment order or constants uniform over all orders.

Lemma 8.1 (Linewise cell count)**.** *A nonaxis line meets at most $2G-1$ open cells of a fixed $G\times G$ grid. Its lattice points have at most two parity types. Consequently there are at most $C_{\mathrm{line}}=4G$ nonempty cell/type pieces on it, uniformly in n, root and direction.*

Proof. Inside the open square a nonaxis line crosses each of the $2(G-1)$ internal grid lines at most once. These crossings divide it into at most $2G-1$ intervals. Convexity prevents re-entry into a cell. Coincident crossings at a vertex only reduce this count. The integer points are $p+kv$, with types alternating between r and $r+\bar{v}$. Thus there are at most $2(2G-1) \le 4G$ pieces. Their index set is determined before the random process is run. □

For an ordinary line $\ell \ni p$ and one such piece E, put

$$H_{\ell,E}(t) = r_\ell(t) \sum_{\substack{q \in (\ell \cap E)\setminus\{p\} \\ q \text{ active}}} w_q(t), \qquad L_{\ell,E} = |(\ell \cap E)\setminus\{p\}|.$$

Stop the fixed-label hierarchy at the first arrival at $H_{\ell,E} \ge L_{\ell,E} + en$, or the root boundary, whichever comes first. As usual, observations at a boundary precede changes of positive-retirement flags. For fixed m, let $\Theta_m(n) = (1-\epsilon_n)^{-\binom{m+1}{2}}$. Use the common-stop form (34) with deterministic horizon T. Expansion by the number a of distinct leaves gives

$$\mathbb{E}^{(p)} H_{\ell,E}^m \le \Theta_m(n) \sum_{a=1}^{m} \left\{ {m \atop a} \right\} B^{m-a} L_{\ell,E}^a. \tag{58}$$

The braces are Stirling numbers of the second kind. For distinct leaves the exact connection is r^a; for repetitions use $r^m \le r^a$, $w^j \le B^{j-1}w$. There are no witnesses and no designated-line connections, so no extra fixed multiplier is needed. Ordered distinct-leaf tuples are counted by $(L)_a \le L^a$.

Lemma 8.2 (Finite height sum)**.** *For $e > 0$, integers $m \ge 3$ and $N \ge 0$,*

$$\sum_{h=1}^{N} 4h \left(\frac{2}{2+eh}\right)^m \le \frac{16}{e^2}\left(\frac{2}{2+e}\right)^{m-3}. \tag{59}$$

Proof. Put $a = e/2$. For $h \ge 1$,

$$h(1+ah)^{-m} \le \frac{(1+a)^{-(m-3)}}{a(1+ah)^2},$$

and

$$(1+ah)^{-2} \le \frac{1}{a}\left[\frac{1}{1+a(h-1)} - \frac{1}{1+ah}\right].$$

Sum the latter telescoping difference. Multiplication by four gives (59), since $4/a^2 = 16/e^2$. □

Apply Markov to (58), sum first over the at most C_{line} pieces of each line, then over directions of height h, using $4h$ directions and $L \le 2n/h$. The all-distinct term is bounded by Lemma 8.2. Bounding the denominator in each repeated term by $(en)^m$ gives the explicit remainder

$$\begin{aligned} R_m(n;e) = 4e^{-m}\Bigg[2\left\{ {m \atop 1} \right\} n(B/n)^{m-1} + 4\left\{ {m \atop 2} \right\}(1+\log n)(B/n)^{m-2} \\ + \sum_{a=3}^{m-1} 2^a \left\{ {m \atop a} \right\}\left(1+\frac{1}{a-2}\right)(B/n)^{m-a}\Bigg]. \end{aligned} \tag{60}$$

Thus the probability of any such first crossing, under the fixed root's point law, is at most

$$C_{\text{line}}\Theta_m(n)\left[\frac{16}{e^2}\left(\frac{2}{2+e}\right)^{m-3} + R_m(n;e)\right]. \tag{61}$$

No union over times, or over all roots, has been used. For sufficiently large fixed m, $R_m(n;e) \to 0$; it is enough that $1-(1-d)(m-1)<0$ and $m>2$.

On the complementary good event, sum the cell bounds after multiplying by their deterministic mask values, then integrate (25). The self point costs a further TB/n. The result is

$$1-r_\ell \le \text{deterministic ordinary-line capacity} + C_{\text{line}}Te + TB/n + O(n^{-1}). \tag{62}$$

Choose $e=(1-\rho-\lambda)/(4TC_{\text{line}})$ when $T>0$. The zero-time case has no births and is trivial. For large n, the last two terms are at most $(1-\rho-\lambda)/4$. Using capacity at most λ, the right side is then strictly less than $1-\rho$, precluding a tag. For any requested positive error, choose m large enough that the first term of (61) is below half that error, and then n large enough for $\Theta_m(n)$ and the remainder. This proves

$$\sup_p \mathbb{P}^{(p)}(\text{tag is an early saving cause}) = o(1). \tag{63}$$

This linewise count also applies to the designated-line clock argument (an axis line has at most $2G$ cell/type pieces). It does not replace the full number of cells in the uniform bounded-variation estimates of Section 2.

Completion of the analytic reduction. The weight-cap estimate (57), tag estimate (63), and global-stop estimate (45) give $\sup_p \mathbb{P}^{(p)}(E_p)=o(1)$. Their events may overlap; a union bound suffices. Proposition 4.3 implies $\mathbb{E}|S_n| \ge (1-o(1))nA_n$, whereas the exact nonnegative loss identity also gives $\mathbb{E}|S_n| \le nA_n$. By Lemma 2.3, $A_n = A + O(n^{-1})$, proving Theorem 1.1.

For fixed $c<A$ the expectation exceeds cn for all large n. If every possible legal output had cardinality smaller than cn, its expectation would be smaller too, a contradiction. Equivalently, with $k=\lceil cn\rceil$ and $|S_n|\le 2n$,

$$\mathbb{P}(|S_n|\ge k) \ge \frac{\mathbb{E}|S_n|-(k-1)}{2n-(k-1)} > 0.$$

This is an existence argument; it does not supply a practical algorithm producing the target density on moderate grids. □

9 The finite input and its geometric checker

This section specifies the 481-phase input and proves the geometric upper bounds computed by its checker. The checker evaluates whole closed root cells and whole angular intervals. Its input is an integer field table and a deletion-rank table; neither sampled risks nor search priorities are acceptance premises. The concrete arithmetic results and execution record are given in Section 10.

9.1 A schedule defined by reverse deletion ranks

Index the four types by $r=0,1,2,3$, with $\operatorname{typ}(p) = (p_x \bmod 2) + 2(p_y \bmod 2)$. The file `field_units.txt` contains 1024 nonnegative integers $u_{r,a,b}$, in that order, with $0\le a,b<16$. Set $G=256$ and write $C_{ij} = (i/G,(i+1)/G)\times(j/G,(j+1)/G)$. Define

$$W_r^{(0)}|_{C_{ij}} = \frac{249}{250\cdot 10^8} u_{r,\lfloor i/16\rfloor,\lfloor j/16\rfloor}. \tag{64}$$

All fields and masks are zero on every fine-grid boundary and outside the square. Thus the coarse table defines the values, but the boundary convention is that of the fine grid.

The file `full_rank.bin` contains $4G^2 = 262144$ little-endian signed 32-bit integers $\nu_{r,i,j}$, in type, first coordinate, second coordinate order. A zero-valued cell has rank -1; every positive cell has a rank in $\{0, \dots, 480\}$. For $0 \le j \le 481$, let

$$W_r^{(j)} = W_r^{(0)} \mathbf{1}_{\{\nu \ge j\}}, \qquad E_r^{(j)} = W_r^{(0)} \mathbf{1}_{\{\nu = j\}} \quad (0 \le j < 481).$$

The indicators here are type-dependent cell indicators. Then $W^{(j)} - W^{(j+1)} = E^{(j)}$, and $W^{(481)} = 0$. For every nonempty rank class set $\delta_j = \max_{r,i,k} E_r^{(j)}|_{C_{ik}} > 0$. Use the phases in decreasing rank order, from 480 to 0. On the phase of duration δ_j, use the mask

$$f_r = E_r^{(j)} / \delta_j. \tag{65}$$

It has rational values in $[0, 1]$. This defines the chronological schedule completely, including its durations and masks.

Lemma 9.1 (Reverse deletion and the common deadline)**.** *The field at the end of the rank-j phase is $W^{(j)}$. Every positive typed cell is used once, and its deadline is the end of its rank's phase. Consequently, a risk bound for $W^{(j)}$ on every rank-j root cell is a bound for that cell's actual common-deadline risk. The terminal field is $W^{(0)}$, and*

$$A = \frac{1}{4} \sum_r \int W_r^{(0)}, \qquad T = \sum_{j=0}^{480} \delta_j.$$

Proof. At the indicated endpoint the phases already run are exactly those with ranks $k \ge j$. Their exposures sum to $\sum_{k=j}^{480} E^{(k)} = W^{(j)}$. This equality holds for all four fields at the same time. A positive typed cell belongs to just one rank class, so its last use is precisely that phase. Finally the classes partition all positive typed cells. There are no overlapping accepted-region areas to count with multiplicity. □

All intermediate fields are componentwise at most $W^{(0)}$. Capacities and squared-chord risk are monotone for nonnegative fields. In particular, a terminal-field capacity bound controls the entire schedule. The rank correspondence itself is checked before any geometric calculation: the program rejects an out-of-range rank, a mismatch between positivity and rank, or an empty requested phase.

9.2 Exact maxima on a closed root cell

Let F be a nonnegative rectangular step field, extended by zero outside the unit square. For a direction $(1, u)$, $u > 0$, put

$$C(v; u) = \int_{\mathbb{R}} F(t, v + ut)\, \mathrm{d}t.$$

For a closed root box $B = [a_0, a_1] \times [b_0, b_1]$, its possible intercepts form the interval $l(u) = b_0 - ua_1 \le v \le h(u) = b_1 - ua_0$. At each field vertex (a, b), combine the signed corner coefficients of its incident rectangles, with exterior values zero. The coefficients for one rectangle of value f, in lower-left, lower-right, upper-left, upper-right order, are $-f, +f, +f, -f$.

Lemma 9.2 (Corner representation)**.** *For $u > 0$,*

$$C(v; u) = \frac{1}{u} \sum_{a,b} k_{a,b} (v - b + ua)_+. \tag{66}$$

Proof. For the indicator of $(a_0, a_1) \times (b_0, b_1)$, integrate $\mathbf{1}_{v+ut>b_0} - \mathbf{1}_{v+ut>b_1}$ over $a_0 < t < a_1$. For the first indicator the result is $[(v + ua_1 - b_0)_+ - (v + ua_0 - b_0)_+]/u$, and similarly for the second. Add the identities with the rectangle weights and combine shared corners. A nonaxis line meets the zeroed grid lines in finitely many points, which do not affect this integral. □

Proposition 9.3 (Exact closed-box envelope)**.** *For fixed $u > 0$, the maximum chord of all roots in B is*

$$U_B(u) = \max\{C(v; u) : v \in \{l(u), h(u)\} \cup \{b - ua : l(u) \le b - ua \le h(u)\}\}. \tag{67}$$

The second set ranges over the field vertices. In particular this maximum is obtainable by exact rational arithmetic at rational u.

Proof. By (66), the chord is continuous and piecewise affine in v, with knots among $b - ua$. The maximum on an interval is at an endpoint or at an included knot. Every possible root intercept belongs to that interval, and every intercept in it is attained by some root of the box. □

The implementation sorts projected vertices, combines equal knots, and accumulates the slope and then the value of the chord numerator. A segment tree returns the largest included knot value; the two interval endpoints are evaluated separately. The zero value and zero slope beyond the last knot are checked. Taking a maximum over all knots, rather than only the root interval, also gives a valid bound simultaneously for every root in the square.

9.3 A rigorous angular-bin bound by vertical dilation

For $\delta \ge 0$, let

$$(D_\delta F)(t, y) = \sup_{|z-y| \le \delta} F(t, z).$$

Lemma 9.4 (Tube domination)**.** *For $(x_0, y_0) \in [0, 1]^2$ and $|u - u_0| \le \delta$,*

$$c_{F,1}((x_0, y_0), u) \le c_{D_\delta F,1}((x_0, y_0), u_0).$$

Proof. Only $0 \le t \le 1$ contributes. On this interval, $|(u - u_0)(t - x_0)| \le \delta$. The point of the original line is therefore among those in the vertical supremum at the corresponding point of the central line. Integrate the resulting pointwise bound. □

Choose $N = 2048$, $H = 2N$, $\delta = 1/H$, and $u_k = (2k + 1)/H$, $0 \le k < N$. Lemma 9.4 gives

$$\sup_{x \in B} \int_0^1 c_{F,1}(x, u)^2 \, \mathrm{d}u \le \frac{1}{N} \sum_{k=0}^{N-1} \left(\sup_{x \in B} c_{D_\delta F,1}(x, u_k) \right)^2. \tag{68}$$

This is a bound on all angles in each bin, not a quadrature approximation whose error has been omitted.

The dilation is itself a finite rational rectangular field. In each column, let the original cell values be f_j, with exterior values zero. Since $\delta < 1/(2G)$, the central strip of cell j has value f_j, and the strip of width 2δ about its boundary j/G has value $\max(f_{j-1}, f_j)$. These strips give an explicit decomposition, including the two outer boundaries. Values on strip endpoints do not affect the positive-slope chord. The program forms the integer type-pair field before this dilation; it does not round each type's field separately.

All dilated vertices have coordinates $(a/H, b/H)$ with integers a, b. At $u = p/H$, their projected intercepts are $Hb - pa$. The integer sweep computes a numerator whose raw-field

chord is that numerator divided by pH. With rounding scale $Q_c = 10^6$, the true scaled chord in (64) is bounded in $1/Q_c$ units by

$$L = \left\lceil \frac{249\,\text{numerator}}{25000\,pH} \right\rceil .$$

This uses $Q_c\,249/(250\cdot 10^8) = 249/25000$. The bin contributes the integer upper bound $\lceil L^2/N \rceil$ in units $1/Q_c^2$. All divisions round upward.

9.4 Complete Farey partitions and rational integral bounds

If the tube estimate is too large to accept a phase, the checker uses the same residual field and every root box of that phase in a second calculation. Its partition is the list of all reduced fractions in $[0,1]$ with denominator at most $G = 256$.

A projected-knot ordering or a knot's membership in the root intercept interval can change only at a slope joining two field or root-box vertices. Those vertices are on the $1/G$ grid, so every such slope in $[0,1]$ has reduced denominator at most G. On an open interval between consecutive fractions, the candidate set in (67) is fixed and every candidate has the form $a + b/u$. Thus $U_B(u)$ is convex as a function of $1/u$. Its endpoint secant in that coordinate bounds it from above. Switches between maximizing candidates require no extra partition. At a positive partition endpoint, continuity of the chord and of the root intercept interval ensures that each one-sided candidate limit is bounded by the endpoint maximum used in the calculation.

Let the consecutive positive fractions be $\alpha = a/b$ and $\beta = c/d$, so $bc - ad = 1$ and $\beta - \alpha = 1/(bd)$. Let A, B be upward-rounded endpoint chord values in units $1/Q_c$. The reciprocal-coordinate secant through these rounded values remains an upper bound for $Q_c U_B$.

Lemma 9.5 (Rational affine integral bound)**.** *An upper bound for* $Q_c^2 \int_\alpha^\beta U_B(u)^2\,\mathrm{d}u$ *is the smaller of* $\max(A^2, B^2)/(bd)$ *and*

$$\begin{cases} (A^2 + AB + B^2)/(3bd), & B \le A, \\ (A'^2 + A'B' + B'^2)/(3bd), & B > A, \end{cases} \tag{69}$$

where

$$A' = A + \frac{(B-A)bc}{(ad+bc)^2}, \qquad B' = B + \frac{(B-A)ad}{(ad+bc)^2}.$$

Proof. Write the reciprocal-coordinate secant as $V(u) = s + t/u$. It is nonnegative and monotone on the interval. If $B \le A$, then $t \ge 0$, so the endpoint secant in u is an upper bound. If $B > A$, then $t < 0$ and V is concave; its tangent at $(\alpha+\beta)/2$ is an upper bound. Substitution and $bc - ad = 1$ give tangent endpoint values A', B'. The integral of the square of an affine function with endpoint values x, y on an interval of length $1/(bd)$ equals $(x^2 + xy + y^2)/(3bd)$. The constant endpoint-maximum bound is valid by monotonicity. Take the smaller of these two valid bounds. □

On the first interval $(0, 1/G)$, every active candidate is bounded and of the form $a + b/u$, hence $b = 0$. Its maximum is constant, so the first positive endpoint suffices. The value at angle zero does not affect the angular integral; it does not replace the axis capacity check below.

For an ordinary vertex $(i/G, j/G)$ at angle p/q, the projection is $qj - pi$. The integer numerator is normalized by pG, and the rounded scaled chord is $\lceil 249\,\text{numerator}/(25000\,pG) \rceil$. The tangent values in (69) are retained as exact rational numbers. Only the final integral

bound is rounded upward to an integer in units $1/Q_c^2$. No logarithmic numerical integration is used.

9.5 Four direction families, integer ranges and acceptance

For each of the six unordered type pairs, the checker transforms the field and root box by each of

$$(x, y), \quad (x, 1-y), \quad (y, x), \quad (y, 1-x).$$

These give the four direction families of (4) and preserve the major-coordinate integration parameter. Type labels are retained. No transpose symmetry of the fields or symmetry of the random process is assumed. Each type receives three pairs and four families.

If their twelve integral bounds sum to the integer J, then

$$\mathcal{R}_r(W^{(j)}; x) \le \frac{19J}{750Q_c^2} \quad \text{throughout the checked root box.} \tag{70}$$

The acceptance test is

$$J \le \left\lfloor \frac{9999 \cdot 750Q_c^2}{190000} \right\rfloor = 39469736842105.$$

It implies $\mathcal{R}_r \le 9999/10000 < 1$. Each output row is identified by type and fine-cell coordinates. The replay driver compares the complete set of rows, and its cardinality, with the rank class to exclude missing or repeated cells. It independently compares the row integers to the rational risk threshold. Passing only selected cells or retaining an unsuccessful tube bound is insufficient.

For this fixed input, the largest single-type integer is 628311458 and the largest type-pair integer is 1181489833. Deletion and dilation cannot increase the latter bound. Its physical chord, in $1/Q_c$ units and rounded upward, is less than 13000000. A corner coefficient has absolute value at most four times the pair maximum, below 2^{33}. Even at the largest supported angular parameter $N = 65536$, projected coordinates are below 2^{35}; a coarse sum-of-absolute-values bound puts sweep intermediates below 2^{86}. Actual chord numerators are below 2^{65} before multiplication by 249. The values and arithmetic used for chord conversion and squaring are therefore within signed 128-bit range. The sum over all bins of one tube integral is at most $13000000^2 + N$; twelve such sums, and their risk numerator after multiplication by 19, are also far below that range. The same bound holds for a Farey integral with the number of intervals in place of N, using the endpoint-maximum bound on each interval.

For the Farey tangent put $D = (ad + bc)^2$. Since $ad, bc \le G^2$, its rational numerator uses

$$A_L = AD + (B - A)bc, \qquad B_L = BD + (B - A)ad,$$

and denominator $3bdD^2$. The bound $A_L, B_L \le 13000000(D + G^2)$ puts $A_L^2 + A_L B_L + B_L^2$ below 2^{117}, and the denominator below 2^{86}. The final ceiling addition also fits. Ordinary integer division and signed 128-bit intermediates suffice for these fixed data. These estimates are part of the soundness argument; runtime undefined-behavior checks provide an additional implementation check. They are not a claim that arbitrary int64 field tables would be safe without a new range bound.

9.6 Capacity: axes, finite directions and the infinite tail

Exchange the two coordinates, transforming the field and root as well, when the second coordinate of the direction is larger in absolute value. Reverse the direction's orientation if

necessary. It then has the form $v = (h, b)$, with $|b| \le h$. Put $q = b/h$; the horizontal-coordinate chord below, divided by $2h$, is exactly the capacity in (3). For a nonaxis slope $q \neq 0$, the chord of a rectangular cell $(A, B) \times (C, D)$ at intercept v is

$$\left[\min\left(B, \max\left(\frac{C-v}{q}, \frac{D-v}{q}\right)\right) - \max\left(A, \min\left(\frac{C-v}{q}, \frac{D-v}{q}\right)\right)\right]_+ .$$

It is continuous and piecewise affine in v, with knots among $C - qA, C - qB, D - qA, D - qB$. Hence the maximum of the summed chord is attained at an intercept through a grid vertex. The independent capacity calculation uses this direct rectangle-intersection formula, not the corner-sweep routine.

Axis directions need a different rule. In an open horizontal strip the chord is its weighted row sum, whereas on a zeroed horizontal boundary it is zero. Check all open-strip values, and use column sums for the vertical direction. Further boundary zeroing on the fine grid cannot increase these capacities: it removes finitely many points from a nonaxis chord and may zero an entire axis chord.

Enumerate all 72 primitive unoriented directions of heights one through seven. A primitive direction has nonzero parity, so it pairs the four types into two unordered pairs; there are 144 finite direction/type-pair checks. The 16 directions of height at most three are designated. For every remaining direction with $h \ge 8$,

$$L_{v,r}(W^{(0)}; x) \le \frac{\|W_r^{(0)} + W_{r+\bar{v}}^{(0)}\|_\infty}{2h} \le \frac{\|W_r^{(0)} + W_{r+\bar{v}}^{(0)}\|_\infty}{16}.$$

Checking the six pair maxima therefore completes the infinite tail. All these calculations may use the coarse 16×16 envelope.

Proposition 9.6 (Finite geometric checking contract)**.** *Suppose the specified field and rank tables satisfy the stated format and positivity correspondence, every rank class is covered by its checked closed cells, and the exact tube or Farey calculation proves the common bound $U_R \le 9999/10000$. Suppose the capacity calculation covers all finite directions and the tail, with bounds $U_H < K$ and $U_L < \lambda < 1 - \rho$. If $9999/10000 < d < 1$, then the schedule (65) has the geometric input properties (5)–(7), with uniform positive margins and the exact mass given by its finite field sum.*

Proof. The capacity calculation covers every line intercept and direction, and terminal-field domination transfers it to all times. Proposition 9.3, Lemma 9.4 and Lemma 9.5 justify the angular bounds on every checked closed cell. Equation (70) gives the risk bound, while Lemma 9.1 identifies the four tested fields with the four actual fields at that root's deadline. The finite maxima leave the positive margins $K - U_H$, $\lambda - U_L$ and $d - 9999/10000$, independent of n. The rank classes partition all positive typed cells, giving the mass without overlap ambiguity. Proposition 2.7 and (22) then provide the uniform discrete comparison used by the stochastic argument. □

The executable `independent/check_phase.cpp`, with `farey_independent.inc`, implements these two angular bounds. Its explicit rectangles, standard sorting, segment tree, ordinary division, wider intermediates and four separate coordinate transforms provide a second implementation of the geometric conditions. Both implementations still use the mathematical lemmas proved here. Input hashes identify the data; they do not prove the inequalities. For a complete reproduction the checker is rebuilt and every phase is evaluated from the fixed input, followed by per-cell coverage and integer comparisons. Reusing cached phase reports without binding them to the current inputs and checker is not such a reproduction.

10 The 481-phase application and a fixed error budget

10.1 One concrete finite input

We use the field and reverse ranks defined in Section 9, with spatial grid $G = 256$, time scale 249/250, and 481 nonempty phases. There are 242,176 positive type/cells; each is used exactly once. By Lemma 9.1, their reverse order defines masks in $[0, 1]$, and the checked background field is the cumulative field at the end of the corresponding forward phase. The four types use a common endpoint. This construction replaces the earlier candidate input; its mass is a direct sum over disjoint type/cells, with no unverified union-area correction.

The exact mass and duration are

$$A = \frac{3842853135627}{2560000000000} = 1.501114506104296875, \qquad T = \frac{30088384743729}{25000000000}. \tag{71}$$

The field is already time-scaled in these formulas. No further factor $(249/250)^2$ is applied to its squared-chord risk. The coefficient of the twelve integrals is $D_*/24 = 19/750$.

Quantity	Exact value or choice
Target c	1501/1000
Designated clock cap K	255/128
Ordinary capacity cap λ	4/5
Ordinary credit floor ρ	1/8
Continuum risk bound r_*	9999/10000
Weight exponent d	19999/20000
Virtual exponent γ	39999/40000
Geometric and correction exponents ξ, ζ	$10^{-6}, 10^{-6}$
Finite-sieve coefficient D_*	76/125
Spatial grid; designated direction count	256; 16
Total early-retirement probability budget	10^{-5}

The capacity bounds for this same field are

$$U_H = \frac{159305594247}{80000000000}, \qquad K - U_H = \frac{69405753}{80000000000},$$
$$U_L = \frac{2535641192953}{3200000000000}, \qquad \lambda - U_L = \frac{24358807047}{3200000000000},$$
$$U_{\text{tail}} = \frac{294190968417}{400000000000}, \qquad 1 - \rho - \lambda = \frac{3}{40}.$$

All margins are strict. The finite calculation covers 144 direction/type-pair rows, including both axes; six tail bounds cover all heights at least eight.

The maximum in the freshly recomputed complete angular-replay records is

$$r_{\text{rec}} = \frac{93740574637267}{93750000000000}, \qquad r_* - r_{\text{rec}} = \frac{50362733}{93750000000000} > 0. \tag{72}$$

It occurs at reverse rank 205, type 2, microcell $(5, 238)$. Every positive type/cell is covered by exactly one phase record. The checking algorithms and their soundness are described in Section 9; the execution status for this edition is stated separately in Section 11.

The finite sieve consists of the 175 odd primes from 3 to 1049, with

$$D_P = \frac{3}{4}\prod_p (1 - p^{-2}) < \frac{76}{125}.$$

For the discrete pair sum set

$$a = r_* + \frac{1}{320000} = \frac{319969}{320000} < d. \tag{73}$$

Equation (22) gives $\mathcal{B}_n^{\Gamma^c} \le a \log n$ once $\log n \ge 3 \cdot 10^{18}$. This explicitly pays the additive lattice error; its coefficient is not silently left equal to r_*. The exponent conditions hold because

$$1 - d - 2\xi - \zeta = \frac{47}{10^6} > 0, \qquad \gamma - d = \frac{1}{40000} > 0.$$

10.2 Three global causes with fixed orders

All moment orders are chosen before taking n to infinity. For the pair-correction and virtual-deletion guards use $m_{\mathrm{guard}} = 2^{18} = 262144$. Including the factor $B = n^d$ for transport to the root point law, the powers in (36) and (37) are respectively

$$4 - m + 2m\xi + d(m-1) + m\zeta + d = -8.320768,$$
$$3 - \gamma m + d(m-1) + \xi + d = -3.553599.$$

Each of these two probabilities is therefore less than $2 \cdot 10^{-6}$, uniformly in the root, for sufficiently large n. Their constants are finite for this fixed order; no unspecified constant is assigned an assumed numerical value.

For designated-line clocks, $C_{\mathrm{line}} = 4G = 1024$ and

$$\varepsilon_H = \frac{K - U_H}{8 C_{\mathrm{line}} T} = \frac{115676255}{1314581588376682496}.$$

Equations (43)–(44) give the same budget $2 \cdot 10^{-6}$. In particular, $\log n \ge 10^7$ suffices for this clock estimate. This is a threshold for one cause, not a value of the final n_0.

10.3 The weight cap with a fixed moment order

By (56), for every fixed integer m, uniformly in the root,

$$I_p \le Y_n + R_n, \qquad \|Y_n\|_{L^m(\mathbb{P}^{(p)})} \le (r_* + o_m(1)) \log n, \qquad \mathbb{E}^{(p)} R_n = o(\log n),$$

where $Y_n, R_n \ge 0$. Put

$$h = \frac{d - r_*}{4} = \frac{1}{80000}, \qquad b = \frac{d + r_*}{2} = \frac{39997}{40000}, \qquad m_{\mathrm{cap}} = 2^{22} = 4194304.$$

After increasing n by a root-independent threshold, we have

$$\|Y_n\|_{m_{\mathrm{cap}}} \le (r_* + h/2) \log n, \qquad \mathbb{E}^{(p)} R_n \le 10^{-6} h \log n. \tag{74}$$

For $q = (r_* + h/2)/(b - h) = 159985/159986$, Bernoulli's inequality gives $q^{-159985} \ge 2$. Since $26 \cdot 159985 \le m_{\mathrm{cap}}$, Markov's inequality yields

$$\mathbb{P}^{(p)}(I_p > b \log n) \le q^{m_{\mathrm{cap}}} + 10^{-6} \le 2^{-26} + 10^{-6} < 2 \cdot 10^{-6}. \tag{75}$$

The two terms pay for $Y_n > (b - h) \log n$ and $R_n > h \log n$ separately.

Both axes are designated directions, so an active root has $\sum_{\ell \ni p,\, \ell \in \Gamma} b_{N_\ell} \ge 1$ in Lemma 4.4. Its self-correction contribution is nonpositive. Each designated factor is at most $2/(2-K) = 256$, and there are 16 designated directions. Thus

$$\log w_p \le I_p + 16 \log 256 < I_p + 128.$$

Reaching $w_p = n^d$ requires $I_p > d \log n - 128$. If $\log n > 5120000$, this implies the event in (75), so the fourth cause fits its budget.

10.4 Ordinary-line tags with a fixed moment order

Use the line-cell allowance

$$e = \frac{1-\rho-\lambda}{4TC_{\text{line}}} = \frac{9765625}{641885541199552}, \qquad b_{\text{tag}} = 131458159, \qquad k_{\text{tag}} = 87,$$

and fix $m_{\text{tag}} = b_{\text{tag}}k_{\text{tag}} + 3 = 11436859836$. Since $b_{\text{tag}}e/2 \geq 1$, Bernoulli's inequality implies

$$\left(\frac{2}{2+e}\right)^{m_{\text{tag}}-3} \leq 2^{-87}.$$

The principal term in (61) is bounded by

$$\frac{16C_{\text{line}}}{e^2}\,2^{-87} = \frac{1005900996096293378894250 49}{219902325555200000000000000000000} < \frac{1}{2000000}. \tag{76}$$

For this fixed order, $m > 2$ and $1-(1-d)(m-1) < 0$; hence the remainder in (60) tends to zero. Increase n until $\Theta_m(n) \leq 2$ and $C_{\text{line}}R_m(n;e) \leq 1/2000000$. The tag probability is then less than

$$2\left[\frac{16C_{\text{line}}}{e^2}2^{-87} + \frac{1}{2000000}\right] < 2\cdot 10^{-6}.$$

The credit identity and (62), with this same e, ensure that a tag cannot hit its floor without a cell crossing. This large moment order is one fixed integer; no expansion into that many monomials is part of the numerical verification.

10.5 The final rational comparison

There are five causes, each with probability less than $2\cdot 10^{-6}$ under the appropriate root point law. The maximum of their finitely many root-uniform thresholds gives

$$\sup_p \mathbb{P}^{(p)}(E_p) < 10^{-5} \tag{77}$$

for every sufficiently large n. The causes need not be independent or disjoint. This is a bound for each root under its own point law, not a simultaneous-success probability for all roots under one law.

Corollary 10.1 (The numerical lower bound)**.** *For every sufficiently large integer n, there is a set $S \subseteq \{0,\dots,n-1\}^2$ with no three distinct collinear points and $|S| \geq \lceil 1501n/1000\rceil$.*

Proof. The finite input satisfies Proposition 9.6, with the values above. The lattice comparison and positive margins supply the hypotheses of the stopped process. Equation (16) gives $A_n \geq A(1-512/n)^2$ for every integer $n \geq 512$. By Proposition 4.3 and (77), $\mathbb{E}|S_n|/n \geq (1-10^{-5})A_n$. For $n \geq 2^{24}$, in addition to the probability thresholds,

$$\begin{aligned}\frac{\mathbb{E}|S_n|}{n} - \frac{1501}{1000} &\geq (1-10^{-5})A(1-2^{-15})^2 - \frac{1501}{1000}\\ &= \frac{2165079963272809251797}{27487790694400000000000000} > 0.\end{aligned} \tag{78}$$

Every output is legal by Proposition 3.1. Some output therefore has cardinality at least $\lceil 1501n/1000\rceil$. All comparisons apply to every sufficiently large integer n, with no parity or divisibility restriction. □

A legal set has at most two points per row, so $|S_n| \le 2n$. Equation (78) also gives

$$\mathbb{P}\{|S_n| \ge \lceil 1501n/1000 \rceil\} > \frac{1}{64000}$$

past the same thresholds: on the complementary event $|S_n| < 1501n/1000$, and the rational margin divided by $2 - 1501/1000$ exceeds $1/64000$. The proof does not compute the final n_0 or provide a practical bound for small-grid simulations.

11 Verification scope and provenance

11.1 Finite computation and mathematical proof

The input consists of 1,024 nonnegative field integers and 262,144 ranks. The latter assign all 242,176 positive type/cells to 481 phases. Source, data and replay records are distributed under `verification/`. The checker evaluates all four direction families directly, without assuming reflection or type symmetry. The 481 freshly accepted records use 385 tube bounds, 95 Farey bounds, and one global tube bound.

Fresh execution for this edition. On 10 November 2026 the checker was rebuilt using GCC 13.3.0 on Ubuntu under WSL, with four OpenMP threads and undefined-behavior sanitization enabled. No cached phase result was reused. All 481 phases and 242,176 positive type/cells passed; coverage checking found no missing, duplicate or failing cell. The complete angular replay took 1,355.4 seconds (about 22.6 minutes). All 481 accepted CSV files are byte-identical to the incoming dossier's records, and the maximum is exactly (72). No sanitizer error was reported. Capacity and clock-constant checks also passed. The separate arithmetic ledger passed 69 checks, including actual field/rank reconstruction and agreement with this run.

The reproduction driver `verification/reproduce.py` requires a new output directory and records the compiler, flags, input and source hashes, executable hash, commands, exit codes and result hashes. Unsafe cached resumption has been removed from the Python wrapper. The mathematical C++ sources and fixed input are unchanged from the reviewed independent implementation. Execution provenance and all per-phase results are included in `verification/evidence/`; the machine-specific executable is rebuilt from source rather than distributed.

The soundness arguments in Section 9 connect finite integer computations to geometric bounds. Hashes establish which files were used; hashes alone do not prove those bounds. Likewise, exact arithmetic certifies finite rational inequalities and field/rank reconstruction, not the stochastic lemmas. Those lemmas are proved in the preceding sections as natural-language mathematics.

The proof uses fixed orders and a fixed error budget. It needs neither growing-order moment estimates nor an explicit value of the final n_0. The available partial Lean development concerns earlier parts of the project and does not formally certify this manuscript or its 481-phase application. No independent human review or complete formal verification is claimed.

11.2 Companion files

The editable manuscript is `main.tex` and its section files. `data/parameters.json` gives exact parameters and finite-sieve factors. `scripts/check_arithmetic.py` reconstructs the input mass, phase durations and supports from the field and rank files, then checks capacity margins, discrete-risk slack, fixed-order witnesses and the final rational budget. Its report is `evidence/arithmetic.json`. Geometric replay, capacity checking and execution provenance

have separate reports in `verification/evidence/`; the arithmetic script does not replace them. Source identifiers and hashes are retained in the companion provenance records.

11.3 AI use and licensing

This work involved substantial use of OpenAI ChatGPT/Codex and AI reviews supplied by the project maintainer. AI systems contributed mathematical proposals and derivations, numerical programs, exact arithmetic checks, Lean definitions and proofs, build orchestration, review, translation and exposition. Separate AI contexts reviewed parts of the work. This English edition was assembled and checked by AI assistance; separate AI reviews are not independent human peer review.

The human maintainer chose the direction and requested experiments and revisions as a hobby project. This disclosure does not attribute every mathematical or programming step to human authorship. No named author, affiliation or personal contact address is supplied. The repository's hosting account is public. The project material and this manuscript are distributed under the MIT license in `LICENSE`; retain its notices when redistributing.

References

[1] Project preparation notes, *Integrated proof draft preceding Lean formalization*, 8–9 November 2026 (Japanese). Fixed-version text. Historical source for the process and stopping arguments.

[2] Project first followup, 9 November 2026 (Japanese). Fixed-version text. Historical source for remaining exposure and the fixed loss budget.

[3] Project second followup, 9 November 2026 (Japanese). Fixed-version text. Historical source for the finite sieve and linewise cell count.

[4] *no-three-in-line-lean*, partial formalization snapshot, 10 November 2026. Source snapshot. This is a partial formalization, not a formal certificate of the present numerical theorem.

[5] Project 481-phase dossier and fixed-budget closure, 10 November 2026. Finite input and executable verification sources are included in this edition. The relevant arguments are developed in English here; the historical Japanese supplements are not required for the mathematical argument.